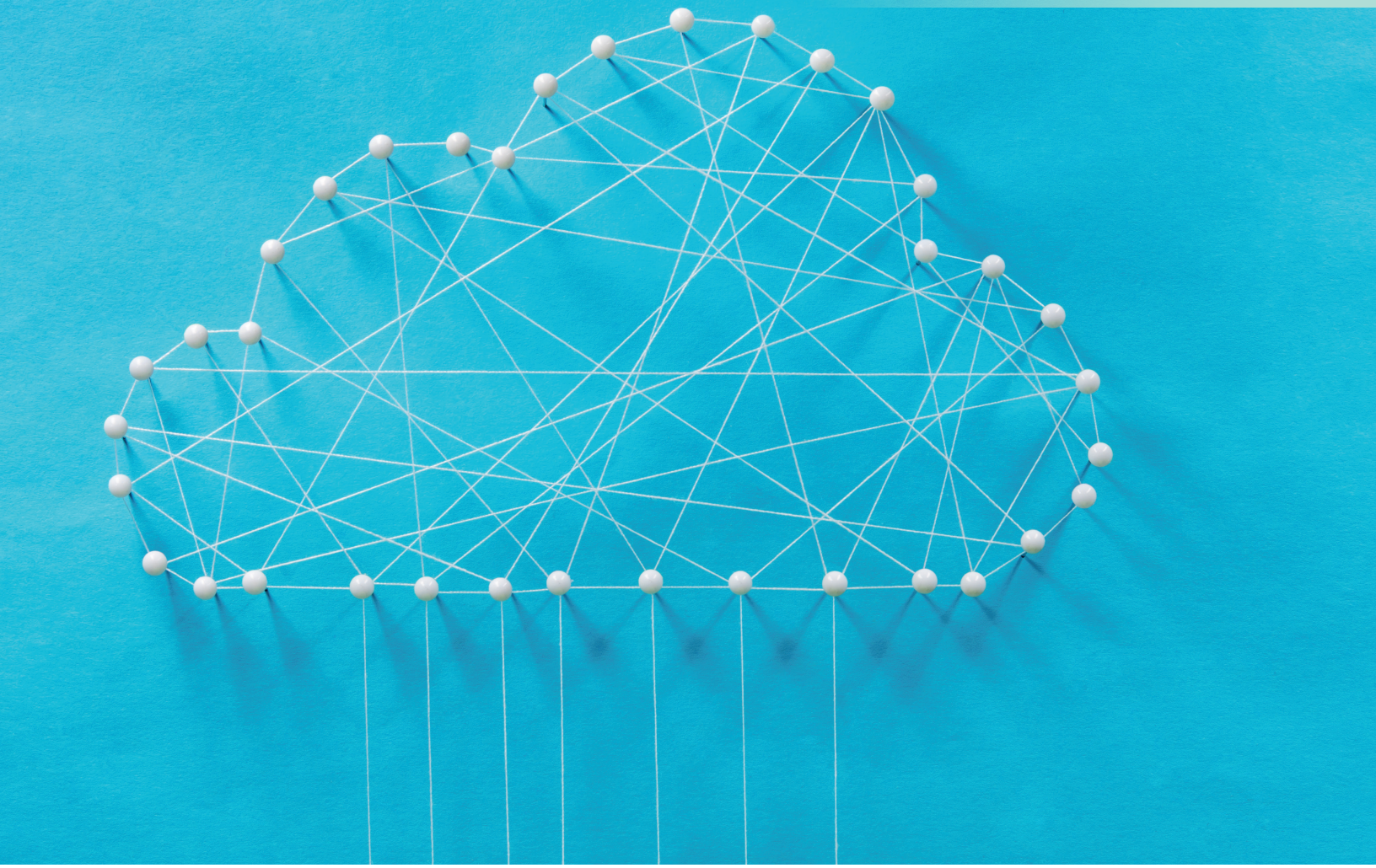
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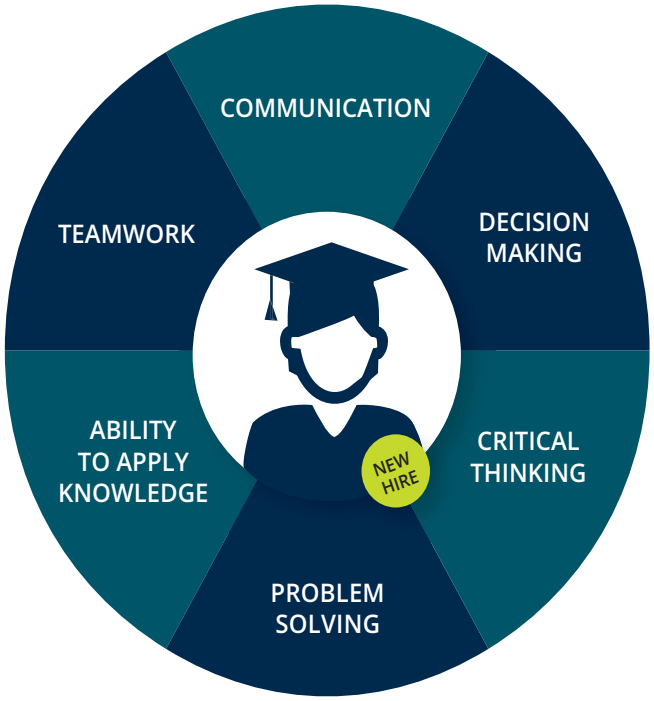
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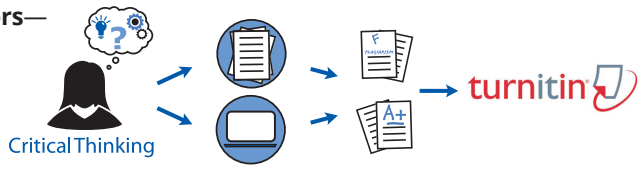
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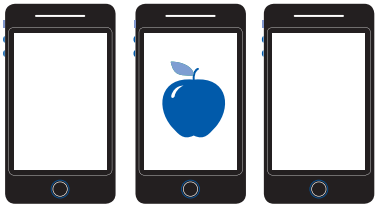
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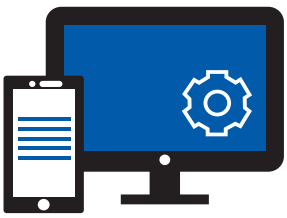
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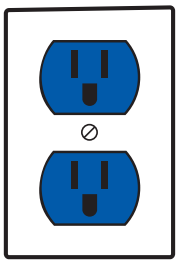
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INFORMATION SYSTEMS TODAY

MANAGING IN THE DIGITAL WORLD

Joseph Valacich
University of Arizona

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City University of Hong Kong



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Dedication

To my mother Mary, you are the best.
—Joe

To Birgit for your love and support.
—Christoph

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About the Authors



Joseph (Joe) Valacich is an *Eller Professor of MIS* within the Eller College of Management at the University of Arizona, a Fellow of the Association for Information Systems (2009), and the Chief Science Officer (CSO) of Neuro-ID, Inc. He was previously on the faculty at Indiana University, Bloomington, and Washington State University, Pullman. He has had visiting faculty appointments at City University of Hong Kong, Buskerud College (Norway), the Helsinki School of Economics and Business, the Norwegian University of Life Sciences, and Riga Technical University (Latvia). He received a PhD degree from the University of Arizona (MIS) and MBA and BS (Computer Science) degrees from the University of Montana. Prior to his academic career, Dr. Valacich worked in the software industry in Seattle in both large and startup organizations.

Dr. Valacich has served on various national task forces designing model curricula for the information systems discipline, including *IS '97*, *IS 2002*, and *IS 2010: The Model Curriculum and Guidelines for Undergraduate Degree Programs in Information Systems,* where he was co-chairperson. He also served on the task force that designed *MSIS 2000* and *2006: The Master of Science in Information Systems Model Curriculum.* He served on the executive committee, funded by the National Science Foundation, to define the *IS Program Accreditation Standards* and served on the board of directors for CSAB (formally the Computing Sciences Accreditation Board) representing the Association for Information Systems (AIS). He was the general conference co-chair for the 2003 International Conference on Information Systems (ICIS) and the 2012 Americas Conference on Information Systems (AMCIS); both were held in Seattle.

Dr. Valacich has conducted numerous corporate training and executive development programs for organizations, including AT&T, Boeing, Dow Chemical, EDS, Exxon, FedEx, General Motors, Microsoft, and Xerox. He has served in a variety of editorial roles within various academic journals and conferences. His primary research interests include human–computer interaction, deception detection, technology-mediated collaboration, mobile and emerging technologies, and e-business. He is a prolific scholar, having published more than 200 scholarly articles in numerous prestigious journals and conferences, including: *MIS Quarterly, Information Systems Research, Management Science, Academy of Management Journal, Journal of MIS, Decision Sciences, Journal of the AIS, Communications of the ACM, Organizational Behavior and Human Decision Processes,* and *Journal of Applied Psychology*. He is a coauthor of the leading textbooks *Modern Systems Analysis and Design* (8th ed.) and *Essentials of Systems Analysis and Design* (6th ed.), both published by Pearson.

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Preface

Approach

Information systems have become *pervasive*. *Mobile devices, social media,* and *cloud computing* have transformed organizations and society. Organizations see the possibilities of the *Internet of Things*, in that not only computers but various sensors, motors, actuators, or even cameras can generate a wealth of potentially useful data. Businesses face unprecedented opportunities, but also challenges, through the ability to utilize *Big Data*. What does all this mean? What are the catalysts of these concepts and of all this change? More important, how can organizations thrive in this dynamic and highly competitive marketplace? The answer to these and many similar questions is that information systems and related information technologies are driving innovation, new business models, and hypercompetition. It is little wonder that teaching an introductory course on information systems has never been more crucial—or more challenging.

One of the greatest challenges that we face in teaching information systems courses is how to keep pace in the classroom with what is happening out in the real world. Being relevant to students while at the same time providing the necessary foundation for understanding the breadth, depth, and complexity of information systems has never been more difficult. We wrote *Information Systems Today,* Eighth Edition, with this overarching goal in mind, to be both rigorous *and* relevant. To accomplish this, we want students not only to learn about information systems but also to clearly understand the importance of information systems for individuals, organizations, and society. Additionally, we do not want to simply spoon-feed students with technical terms and the history of information systems. Instead, students must understand exactly what innovative organizations are doing with contemporary information systems and, more important, where things are heading. Finally, we want to empower students with the essential knowledge needed to be successful in the use and understanding of information systems in their careers.

To this end, we wrote *Information Systems Today,* Eighth Edition, so that it is contemporary, fun to read, and useful, focusing on what business students need to know about information systems to survive and thrive in the digital world.

Audience

Information Systems Today, Eighth Edition, is primarily for the undergraduate introductory information systems course required of all business students. The introductory information systems course typically has a diverse audience of students majoring in many different areas, such as accounting, economics, finance, marketing, general management, human resource management, production and operations, international business, entrepreneurship, and information systems. This book was also written for students studying topics outside of business, especially in the growing and broad area of information sciences. Given the range of students taking this type of course, we have written this book so that it is a valuable guide to all students, providing them with the essential information they need to know. Therefore, this book has been written to appeal to a diverse audience.

Information Systems Today, Eighth Edition, can also be used for the introductory course offered at the graduate level—for example, in the first year of an MBA program. Such usage would be especially appropriate if the course heavily focused on the diverse set of cases provided in each chapter.

What's New to the Eighth Edition

Our primary goal for *Information Systems Today,* Eighth Edition, was to emphasize the importance of information systems to all business students as the role of information technology and systems continues to expand within organizations and society. Most notably, we extensively

examine how five big megatrends—mobile, social media, the Internet of Things, cloud computing, and Big Data—are transforming individuals, organizations, and society. Given this clear focus, we are better able to identify those topics most critical to students and future business professionals. Consequently, we have made substantial revisions to the basic content of the chapters and pedagogical elements as well as introduced several new elements that we believe help achieve this goal. New or expanded chapter topics include the following:

- An extensively revised chapter—Chapter 1, "Managing in the Digital World"—focuses not only on defining what an information system consists of but also provides new content on globalization and societal issues in the digital world as well as the role of five IT megatrends in fueling and addressing these issues.
- An extensively revised chapter—Chapter 2, "Gaining Competitive Advantage Through Information Systems"—provides new content describing how information systems play a key part in enabling different types of innovation and innovative business models.
- A revised chapter—Chapter 3, "Managing the Information Systems Infrastructure and Services"—provides updated content on the need for a reliable, adaptable, and scalable infrastructure to support the needs of today's organizations as well as on essential infrastructure concepts related to hardware, software, storage, networking and the Internet, data centers, and cloud computing.
- A revised chapter—Chapter 4, "Enabling Business-to-Consumer Electronic Commerce"—provides updated content related to e-commerce involving the end consumer as well as new and expanded coverage of e-finance, fintech, and related issues.
- A revised chapter—Chapter 5, "Enhancing Organizational Communication and Collaboration Using Social Media"—centers around various topics related to the need for organizational communication and provides updated content on how individuals and organizations use both traditional communication and collaboration tools and social media for communication, collaboration, cooperation, and connection.
- An extensively revised chapter—Chapter 6, "Enhancing Business Intelligence Using Big Data and Analytics"—provides extended coverage on business intelligence and advanced analytics and greatly expanded content on machine learning, predictive modeling, artificial intelligence, unstructured data analytics, and spatial decision support.
- A revised chapter—Chapter 8, "Strengthening Business-to-Business Relationships via Supply Chain and Customer Relationship Management"—provides updated content on business-to-business electronic commerce and supply chain management as well as customer relationship management (CRM).
- A revised chapter—Chapter 9, "Developing and Acquiring Information Systems"—provides updates to various topics and extended content on alternative system development methodologies.
- A revised chapter—Chapter 10, "Securing Information Systems"—provides an update to all topics and deeper coverage on industrial espionage and cyberterrorism.
- A revised Technology Briefing covers foundational concepts related to various information technologies. The Technology Briefing provides the foundations for a deeper understanding of the topics introduced in Chapter 3 and is intended for use in more technically oriented courses. Each section of this briefing was designed to stand alone—it can be read with or without the other sections.

In addition to the changes within the main chapter content, we have also added two new features to each chapter—Green IT and Security Matters. Green IT presents environmental issues arising from the use of information systems. For example, in Chapter 4, we discuss the environmental impacts of online shopping. Security Matters presents some current issues and threats arising from the ubiquitous use of information systems. For example, in Chapter 5, we discuss how hacktivists challenged the extramarital dating website Ashley Madison.

Beyond the chapter content and features, we have also made substantial changes and refinements to the end of each chapter. In particular, we carefully revised many of the end-of-chapter problems and exercises to reflect content changes and new material. Further, we have carefully updated the end-of-chapter cases about contemporary organizations and issues to illustrate the complexities of the digital world. Each case mirrors the primary content of its chapter to better emphasize its relevancy within the context of a real organization. All these elements are discussed more thoroughly next.

Our goal has always been to provide only the information that is relevant to all business students, nothing more and nothing less. We believe that we have again achieved this goal with *Information Systems Today,* Eighth Edition. We hope you agree.

Key Features

As authors, teachers, developers, and managers of information systems, we understand that in order for students to best learn about information systems with this book, they must be motivated to learn. To this end, we have included a number of unique features to help students quickly and easily assess the true value of information systems and their impact on everyday life. We show how today's professionals are using information systems to help modern organizations become more efficient and competitive. Our focus is on the application of technology to real-world, contemporary situations. Next, we describe each of the features that contribute to that focus.

Pedagogy—A Multitiered Approach

Each chapter provides a list of learning objectives to lay the foundation for the chapter content, followed by an opening case to highlight how contemporary organizations are utilizing information systems to gain competitive advantage, streamline organizational processes, or improve customer relationships or how information systems fuel societal change. In addition, throughout each chapter, various short pedagogical elements are presented to highlight key information systems issues and concepts in a variety of contexts. These elements help to show students the broader organizational and societal implications of various topics. At the end of each chapter, the Key Points Review repeats the learning objectives and describes how each objective was achieved; a variety of questions and exercises helps students assess their understanding of the chapter material and encourages them to synthesize and apply the concepts learned. A list of references appears at the end of each chapter.

OPENING CASE—MANAGING IN THE DIGITAL WORLD. Each chapter begins with an opening case describing a real-world company, technology, and/or issue to spark students' interest in the chapter topic. We have chosen engaging cases that relate to students' interests and concerns by highlighting why information systems have become central for managing in the digital world. Each opening case includes a series of associated questions the students will be able to answer after reading the chapter contents. The organizations, technologies, or issues highlighted in these cases are as follows:

- The rise of open innovation
- How information systems fuel startups and new business models
- Google's meteoric rise and its transition to Alphabet
- How Chinese e-commerce company Taobao became a leader in the world of e-commerce
- How Facebook has emerged as one of the most successful and powerful social media sites
- Intelligence through drones
- Amazon.com's use of its sophisticated infrastructure to automate the supply chain for both large and small customers
- How Walmart became a leader in managing its global supply chains
- The rise of the maker movement
- How the hacking group "Anonymous" uses various tactics to further its ideological goals

Green IT Case

Climate change and resource scarcity are among the most pressing issues societies face. To highlight the role of information systems in this context, each chapter includes a Green IT case. This new feature discusses important issues related to the environmental impacts of information systems as well as how information systems can be used to reduce negative environmental impacts. The Green IT cases are embedded in the text of the chapter and highlight concepts from the surrounding chapter material. The issues and organizations highlighted in these cases are as follows:

- Green IT and the Internet of Things
- How the U.S. Navy is using alternative energy sources to address power consumption of its fleets

- How Alphabet uses renewably energy to power its data centers
- The environmental impacts of online shopping
- How green IT is fueling the use of renewable energy
- How the Internet of Things, Big Data, and analytics fuel greener facilities
- Why your ERP system should be in the cloud
- How Nike builds a greener supply chain
- How companies are trying to reduce the carbon footprint of modern data centers
- How Anonymous protests the killing of dolphins and whales in Japan

Security Matters

With information systems becoming ever more ubiquitous, security is of growing concern, not only for organizations but also for individuals. While we dedicate an entire chapter to issues surrounding securing information systems, this new feature presents some current issues and threats. The topics discussed in this element are as follows:

- How computer criminals use ransomware to extort money from organizations and everyday people
- How attackers use the SWIFT system to conduct virtual bank robberies
- How attackers can remotely hack into a car's onboard systems
- How even small companies are not immune from being targeted
- How terrorism is winning the social media battle
- How hacktivists challenged the extramarital dating website Ashley Madison
- How companies have to weigh the benefits and dangers of not updating ERP systems
- How VTech's attackers disclosed the customer data of the most vulnerable
- How attackers use mobile malware to steal online banking users' login credentials
- How analog may be the future of securing critical infrastructure

Coming Attractions

We worked to ensure that this book is contemporary. We cover literally hundreds of different current and emerging technologies throughout the book. This feature, however, focuses on innovations that are likely to soon have an impact on organizations or society. The topics discussed are as follows:

- Storing the history of humankind in memory crystals
- CITE—a city-sized test lab for innovations
- Extending the human lifetime indefinitely
- Using artificial intelligence to manage hedge funds
- Dissolvable electronics to fight bacteria
- Emotion aware gaming
- Transforming ERP and organizations using the Internet of Things
- Reducing supply chain problems using augmented reality
- Harvesting human energy
- Using brainwaves to verify people's identities

When Things Go Wrong

Textbooks don't usually describe what not to do, but this can be very helpful to students. This feature enables students to learn about a real-world situation in which information systems did not work or were not built or used well. The topics and issues discussed are as follows:

- The negative effects of technology addiction
- The pains of Uber in China
- Dirty data centers and the environmental impact of cloud computing
- How companies are trying to rig "likes" to gain reputation on social networking sites
- Crowdfunding failures
- How Twitter can quickly disseminate *mis*information, with unforeseen consequences

- How a software error freed prisoners early
- How a supply chain failure caused SpaceX rocket failure
- Top security threats
- How the "heartbleed" bug almost killed the Internet

Who's Going Mobile

Mobile technologies have become pervasive throughout society. New opportunities and issues have emerged with the growing importance of mobile devices, such as smartphones and tablets, which are in people's immediate reach 24/7. Related to each chapter's content, this feature examines topics related to the growth in mobile device usage throughout the world. The topics discussed are as follows:

- The rise of wearable technologies
- How information systems support the lifestyle of the digital nomads
- How mobile payment systems have transformed developing countries
- The rise of mobile payments
- Going SoLoMo: Yelp
- Identifying malaria hotspots using mobile phone data
- Managing businesses on the road using mobile ERP
- Developing mobile CRM apps for customers
- How to succeed in mobile app development
- Backdoors in mobile phones

Ethical Dilemma

Ethical business practices are now a predominant part of contemporary management education and practice. This feature examines contemporary dilemmas related to the chapter content and highlights the implications of these dilemmas for managers, organizations, and society. Discussion questions are provided to seed critical thinking assignments or class discussions. The topics discussed are as follows:

- The social and environmental costs of the newest gadgets
- The ethics of the sharing economy
- The ethics of publishing street photography on the web
- The ethics of reputation management
- Anonymity, trolling, and cyberharassment
- The Orwellian Internet of Things
- Privacy issues of radio frequency identification
- Using CRM systems to target or exploit consumers
- Ethical app development
- The ethics of cyberwar

Industry Analysis

Every industry is being transformed by the Internet and the increasing use of information systems by individuals and organizations. To give students a feel for just how pervasive and profound these changes are, each chapter presents an analysis of a specific industry to highlight the new rules for operating in the digital world. Given that no industry or profession is immune from these changes, each Industry Analysis highlights the importance of understanding information systems for *every* business student, not only for information systems majors. Discussion questions help students better understand the rapidly changing opportunities and risks of operating in the digital world. Chapter 1 examines how the digital world is transforming the opportunities for virtually all business professions. Subsequent chapters examine how globalization and the digital world have forever transformed various industries, including education, entertainment, retail, travel, health care, automobile, manufacturing, broadcasting, and law enforcement. Clearly, we are in a time of tremendous change, and understanding this evolution will better equip students to not only survive but also thrive in the digital world.

End-of-Chapter Material

Our end-of-chapter material is designed to accommodate various teaching and learning styles. It promotes learning beyond the book and the classroom. Elements include the following:

- ***Key Terms***—Highlight key concepts within the chapter.
- ***Review Questions***—Test students' understanding of basic content.
- ***Self-Study Questions***—Enable students to assess whether they are ready for a test.
- ***Problems and Exercises***—Push students deeper into the material and encourage them to synthesize and apply it.
- ***Application Exercises***—Challenge students to solve two real-world management problems using spreadsheet and database applications from a running case centered on a university travel agency. Student data files referenced within the exercises are available on the book's website: www.pearsonglobaleditions.com/valacich.
- ***Team Work Exercise***—Encourage students to keep up with, discuss, visualize, and present interesting, important trends and forecasts related to Internet usage within a variety of contexts.

We have extensively updated these elements to reflect new chapter content and the natural evolution of the material.

End-of-Chapter Cases

To test and reinforce chapter content, we present two current real-world cases at the end of each chapter. Like the Opening Cases of each chapter, these cases are taken from the news and are contemporary. However, these are longer and more substantive than the Opening Cases. Sources for these cases include *BusinessWeek, CIO* magazine, *InformationWeek, Wired,* and various websites. They too are followed by discussion questions that help the student apply and master the chapter content. The organizations, products, and issues highlighted in these cases are as follows:

- Apple's rise, fall, and reemergence as a global technology giant
- How electronic health records are transforming healthcare
- How LinkedIn provided a professional networking platform through which users could consume content as well as create value
- How streaming video is disrupting the movie rental and TV broadcasting industries
- How Amazon Web Services transformed itself and became a pioneer in the use of cloud computing in Asia
- How the dark web fuels illegal activities
- How web analytics are providing unprecedented insights into online consumer behavior
- How Rocket Internet aims to become a European Internet giant by cloning business models
- How algorithms determine news feeds
- How scammers use like farming and clickbait to game Facebook's newsfeed algorithms
- How healthcare in Singapore has used big data and analytics
- How companies gather social intelligence through social media
- How software as a service has enabled small and medium-sized organizations to utilize enterprise resource planning (ERP) systems
- How TH Milk Vietnam has adopted a cloud-based ERP system
- How natural disasters disrupt global supply chains
- How companies attempt to use information systems to efficiently deliver products to the "last mile"
- How the Federal Bureau of Investigation and Department of Homeland Security joined forces in developing a comprehensive database of biometric information to better track and apprehend criminals
- How Hadoop and MapReduce fuel the use and analysis of Big Data
- How SingPass gives the residents of Singapore access to e-government services
- How China limits information exchange within its society through its "great firewall"

Organization

The content and organization of this book are based on our own teaching as well as on feedback from reviewers and colleagues throughout the field. Each chapter builds on the others to reinforce key concepts and allow for a seamless learning experience. Essentially, the book has been structured to answer three fundamental questions:

1. What are contemporary information systems, and how are they being used in innovative ways?
2. Why are information systems so important and interesting?
3. How best can we build, acquire, manage, and safeguard information systems?

The ordering and content of our chapters were also significantly influenced by the "IS 2010 Curriculum Guidelines for Undergraduate Degree Programs in Information Systems"[1]; these guidelines, written by prominent information systems scholars, define the information systems core body of knowledge for all business students. By design, the content of *Information Systems Today,* Eighth Edition, carefully follows these guidelines, and we are, therefore, very confident that our book provides a solid and widely agreed-on foundation for any introductory information systems course.

The chapters are organized as follows:

- ***Chapter 1: Managing in the Digital World***—Information systems are fueling change in the digital world. Here, we help students understand what information systems are, the pressing issues societies in the digital world are facing, how five IT megatrends—mobile, social media, the Internet of Things, cloud computing, and Big Data—influence organizations and society, and how information systems have become a vital part of modern organizations. We walk the student through the technology, people, and organizational components of an information system, and lay out types of jobs and career opportunities in information systems and in related fields. We also focus on how technology is creating countless ethical concerns.
- ***Chapter 2: Gaining Competitive Advantage Through Information Systems***—Given the rapid advancement of new technologies, we explain why and how companies are continually looking for innovative ways to use information systems for competitive advantage, and how information systems support organizations' business strategies. Here, we discuss how companies from GE to Uber can use information systems for automation, organizational learning, and strategic advantage by creating new and innovative business models.
- ***Chapter 3: Managing the Information Systems Infrastructure and Services***—With the ever-increasing complexity of maintaining a solid information systems infrastructure, it becomes increasingly important for organizations such as Google to design a reliable, robust, and secure infrastructure. Here, we provide an overview of the essential information systems infrastructure components and describe why they are necessary for satisfying an organization's informational needs. We also examine the rapid evolution toward the delivery of infrastructure capabilities through a variety of cloud-based services.
- ***Chapter 4: Enabling Business-to-Consumer Electronic Commerce***—Perhaps nothing has changed the landscape of business more than the use of the Internet for electronic commerce. Here, we describe how firms such as Amazon.com, Dell, or Taobao; governments; financial services providers; and e-finance startups use the Internet to conduct commerce in cyberspace. Further, we describe the requirements for successful e-commerce websites and discuss Internet marketing and mobile commerce as well as consumer-to-consumer and consumer-to-business e-commerce. Finally, we discuss payment and legal issues in e-commerce.
- ***Chapter 5: Enhancing Organizational Communication and Collaboration Using Social Media***—Social media have forever changed how people interact. In addition to enabling various business opportunities, social media have also enabled companies to better harness the power and creativity of their workforce. Here, we provide an overview of traditional communication and collaboration tools and examine how different social media can

[1] Topi, H., Valacich, J., Wright, R. T., Kaiser, K., Nunamaker Jr., J. F., Sipior, J. C., & de Vreede, G. J. (2010). IS 2010: Curriculum guidelines for undergraduate degree programs in information systems. *Communications of the Association for Information Systems, 26*(18).

enhance communication, collaboration, cooperation, and connection within organizations but also between organizations and their customers. Further, we discuss the importance of carefully managing the use of social media within organizations. Finally, using examples such as Twitter and Facebook, we describe how companies can deal with potential pitfalls associated with social media.

- ***Chapter 6: Enhancing Business Intelligence Using Big Data and Analytics***—A key to effective management in a global, highly competitive, and rapidly changing environment is high-quality and timely information to support decision making in order to realize the strategic goals of the organization. Here, we first describe the need for enhanced decision making and explain how databases serve as a foundation for gaining business intelligence. We then discuss concepts related to business intelligence and advanced analytics, including data mining, machine learning, and predictive modeling. Finally, we discuss how knowledge management and geographic information systems help organizations make better business decisions.
- ***Chapter 7: Enhancing Business Processes Using Enterprise Information Systems***—Enterprise systems have become a critical technology in a broad range of organizations, both large and small, to integrate information and span organizations' boundaries to better connect a firm with customers, suppliers, and other partners. Here, we focus on foundational concepts related to enterprise systems, walking students through various core business processes, and then examine how enterprise resource planning systems can be applied to improve these processes and organizational performance.
- ***Chapter 8: Strengthening Business-to-Business Relationships via Supply Chain and Customer Relationship Management***—Two additional types of enterprise systems, supply chain management systems and customer relationship management systems, are being used to facilitate various business processes between suppliers and customers. Here, we begin by introducing business-to-business electronic commerce. Next, we examine how supply chain management systems can support the effective management of supply networks. Finally, we examine customer relationship management systems and their role in attracting and retaining customers and, using examples from companies such as Dell, discuss how organizations can integrate social media in their CRM efforts.
- ***Chapter 9: Developing and Acquiring Information Systems***—Nearly every organization needs to develop or acquire information systems. Here, we begin by describing how to formulate and present the business case to build or acquire a new information system. We then walk the student through the traditional systems development approach and explain how numerous other approaches, such as agile development, can be utilized depending on the situation. Finally, we examine the steps followed when acquiring an information system from an outside vendor.
- ***Chapter 10: Securing Information Systems***—With the pervasive use of information systems, new dangers have arisen for organizations, and the interplay between threats, vulnerabilities, and potential impacts has become a paramount issue within the context of global information management. Here, we contrast several types of computer crime and discuss the growing significance of cyberwar and cyberterrorism. We then highlight the primary threats to information systems security and explain how systems can be compromised and safeguarded. We conclude this chapter with a discussion of the role of auditing, information systems controls, and the Sarbanes–Oxley Act. Note that some instructors may choose to introduce this chapter prior to the discussion of the information systems infrastructure in Chapter 3.
- ***Technology Briefing***—In addition to these 10 chapters, we include a Technology Briefing that focuses on foundational concepts regarding hardware, software, networking and the Internet, and databases. While Chapter 3, "Managing the Information Systems Infrastructure and Services," provides a more managerial focus to these enabling technologies, this foundational material provides a more in-depth examination of these topics. By delivering this material as a Technology Briefing, we provide instructors the greatest flexibility in how and when they can apply it.

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Instructor Resources

At the Instructor Resource Center, www.pearsonglobaleditions.com/valacich, instructors can easily register to gain access to a variety of instructor resources available with this text in downloadable format. If assistance is needed, our dedicated technical support team is ready to help with the media supplements that accompany this text. Visit http://support.pearson.com/getsupport for answers to frequently asked questions and toll-free user support phone numbers.

The following supplements are available with this text:

- Instructor's Resource Manual
- Test Bank
- TestGen® Computerized Test Bank
- PowerPoint Presentation
- Image Library

Reviewers

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1 Managing in the Digital World

Preview

Today, organizations from Apple to Zappos use information systems to better manage their organizations in the digital world. These organizations use information systems to provide high-quality goods and services as well as to gain or sustain competitive advantage over rivals. In addition to helping organizations to be competitive, information systems have contributed to tremendous societal changes. Our objective for this chapter is to help you understand the role of information systems as we continue to move further into the digital world, the role of information systems in current issues faced by societies in the digital world, and the role of information technology (IT) megatrends in influencing the digital future. We then highlight what information systems are, how they have evolved to become a vital part of modern organizations, and why this understanding is necessary for you to become an effective manager in the digital world. We conclude by discussing ethical issues associated with the use of information systems.

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MANAGING IN THE DIGITAL WORLD:

Open Innovation

Where do good ideas come from? An eccentric inventor toiling alone? A secretive lab filled with researchers in white coats? Views of innovation are shifting away from these traditional stereotypes. For decades, corporations funded internal research and development units and tightly controlled both the inputs and outputs of these operations. Opportunities to interact with customers were limited, and the possibility of spending months or years and millions of dollars developing products that no one wanted was a real threat. New technologies are enabling a shift in the way innovation occurs.

Traditionally, universities would conduct basic and applied research, but the results of this research only sometimes would make their way to the private sector. Corporations would fund their own research and development operations, often at great expense. Such operations took years to set up and were often highly constrained in the types of research they could carry out. Programs of research were evaluated against business plans that had been studied, reviewed, and approved by multiple layers of management. The time and complexity involved in these bureaucratic processes often left the actual research out of date and out of touch with the realities of the marketplace and actual customer wants and needs. The resulting products

After reading this chapter, you will be able to do the following:

1. Describe the characteristics of the digital world, contemporary societal issues of the digital world, and IT megatrends shaping the digital future.
2. Explain what an information system is, contrasting its data, technology, people, and organizational components.
3. Describe the dual nature of information systems in the success and failure of modern organizations.
4. Describe how computer ethics affect the use of information systems and discuss the ethical concerns associated with information privacy and intellectual property.

would often fail in the market due to being years late or no longer being relevant.

Open innovation is a new approach. Instead of relying on tightly controlled internal research projects, companies are opening up their research and development efforts to a broad audience (Figure 1.1). Customers, suppliers, and other companies are invited to participate more directly in different phases of the innovation process, and companies are working more collaboratively with universities.

Many companies take these ideas even further and open up the research and development efforts to anyone who wishes to participate online or in person. For example, Starbucks introduced "My Starbucks Idea," where customers can post ideas and suggestions as well as vote on or discuss others' ideas. Hundreds of customer-generated ideas have been launched over the years. Likewise, more than 37,000 ideas have been submitted to Dell's "IdeaStorm" website, with more than 550 ideas implemented, and Heineken's "Innovators Brewhouse" uses open innovation to generate ideas related to topics ranging from methods for counterfeit detection to brewing closer to the consumer or new, more convenient packaging. Further, new tools like interactive 3D visualization and rapid prototyping technologies like 3D printing allow for tremendously lowered barriers to entry to innovation. Many companies and institutions have set up collaborative spaces to share resources and encourage the fusion of ideas and skills that can lead to exciting breakthroughs. As with many innovations themselves, this innovative way of innovating would not be possible without information systems.

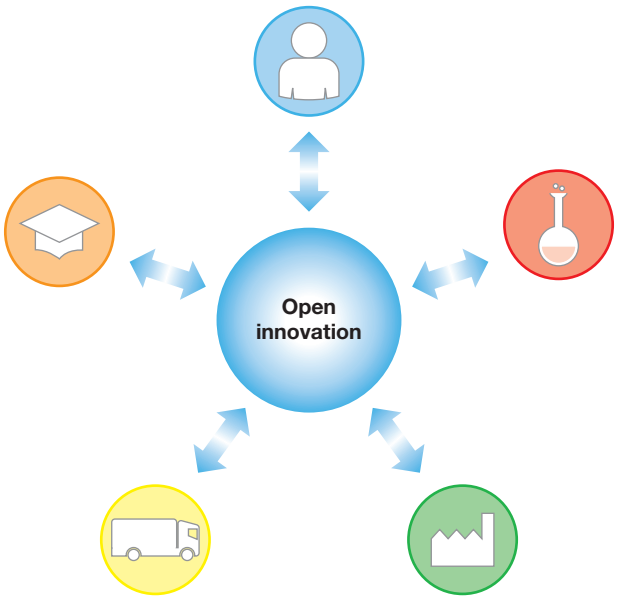


FIGURE 1.1

Open innovation entails opening up the innovation process to outside entities, including academia, individual innovators, research labs, other companies, or suppliers.

After reading this chapter, you will be able to answer the following:

1. How do the five IT megatrends fuel open innovation?
2. What are the primary information systems components that enable open innovation?
3. What intellectual property issues arise from engaging in open innovation?

Based on:

Board of Innovation. (n.d.). List of open innovation and crowdsourcing examples. Retrieved June 20, 2016, from http://www.boardofinnovation.com/list-open-innovation-crowdsourcing-examples

GE. (2016). GE open innovation. *GE.com*. Retrieved April 24, 2016, from http://www.ge.com/about-us/openinnovation

Open Innovation Community. (2016). Open innovation. Retrieved April 24, 2016, from http://openinnovation.net/about-2/open-innovation-definition

Information Systems Today

Today, information systems (IS) are ubiquitous: Be it traditional desktop computers, laptop computers, smartphones, tablets, you name it; information systems are all around us, whether you see them or not. Companies such as FedEx and UPS use information systems to route trucks and track packages. Retailers such as Walgreens and Walmart use information systems for everything from optimizing supply chains to recording purchases and analyzing customer tastes and preferences. Cities use information systems for adaptive traffic control systems or variable speed limits. Cars use information systems for everything from ignition control to airbags to distance control and park assist systems. Many innovative business models, ranging from Airbnb to Uber, are built on or around information systems. Alternatively, just look around your school or place of work. At your school, you register for classes online; use e-mail, Twitter, or Facebook to communicate with fellow students and your instructors; access e-books from your library; and complete or submit assignments on online learning platforms such as Blackboard, Moodle, Canvas, or Sakai. At work, you may use a PC for e-mail and many other tasks. Your paychecks are probably generated by computer and automatically deposited into your bank account via high-speed networks. Even in your spare time, information systems are ubiquitous: You use social networking sites like Facebook to stay connected with your friends and family, you watch videos on YouTube, you upload pictures taken with your smartphone to picture-sharing sites like Instagram, you listen to music on Pandora or Spotify, and you use your smartphone for playing games, sending e-mails, or reading books. Chances are that each year you see more information systems than you did the year before, and these systems are a more fundamental and important part of your social, academic, and work life than ever before.

The Emergence of the Digital World

Over the past decades, the advent of powerful, relatively inexpensive, easy-to-use computers has had a major impact on business and society. When you stop and think about it, it is easy to see why information systems are important. Increasing global competitiveness has forced companies to find ways to be better and to do things less expensively. The answer for many firms continues to be to use information systems to do things better, faster, and cheaper. Many organizations use information systems to support innovative business models, or build their entire business models around technological innovations. Likewise, using global telecommunications networks, companies can more easily integrate their operations to access new markets for their products and services as well as access a large pool of talented labor in countries with lower wages.

Clearly, we are living in a digital world. Given the proliferation of mobile devices such as tablets or smartphones, some have even argued that we are living in the post-PC era, where wireless, mobile devices will replace traditional desktop and laptop computers. In fact, already in the last quarter of 2011, Apple sold more iPads than HP (traditionally one of the world's leading PC makers) sold PCs, and in the United States, smartphone penetration has reached 82 percent (Nielsen, 2016). Initially created as consumer devices, tablets have become commonplace in various professional settings, including warehouses, showrooms, airplane cockpits, and hospitals (Figure 1.2).

Yet desktop PCs and laptops are unlikely to go away. Rather, devices with newer form factors will work in tandem with older form factors to provide truly ubiquitous experiences; mobile devices complement traditional computers, providing different devices for different users and different tasks, where not the device but the services and data provided are of primary importance. Further, the changes we've seen so far have given rise to developments such as wearable computers, augmented reality devices, or surface computers.

Changes in technology have enabled new ways of working and socializing; whereas traditionally, people were bound to a stationary PC to do essential tasks, they can now perform such tasks from almost anywhere they have a cell phone signal. Likewise, workdays traditionally had a clear beginning and a clear end—from when you powered your computer on to when you turned it off at night. Today, many tasks (especially more casual tasks such as reading or sending e-mails) can be done at any time, often in small chunks in between other tasks, such as when waiting in line at the supermarket cashier.

Computing has changed from an activity primarily focused on automating work to encompass various social and casual activities. Devices such as smartphones or tablets, paired with mobile broadband networks, allow for instant-on computing experiences, whenever and



FIGURE 1.2

Mobile devices are increasingly being used in various professional settings.
Source: William Perugini/Shutterstock.

wherever; advances in *cloud computing* (think Gmail, Office Online, or Dropbox) allow for accessing e-mails, files, notes, and the like, from different devices, further enhancing portability and mobility.

In effect, we are in a virtuous cycle (or in a vicious cycle, considering the creep of work life into people's leisure time and the increasing fixation on being permanently "on call"), where changes in technology lead to social changes and social changes shape technological changes. For example, communication, social networking, and online investing almost necessitate mobility and connectivity, as people have grown accustomed to checking e-mails, posting status updates, or checking on real-time stock quotes while on the go. In addition, the boundaries between work and leisure time are blurring, so that employees increasingly demand devices that can support both and often bring their own devices into the workplace.

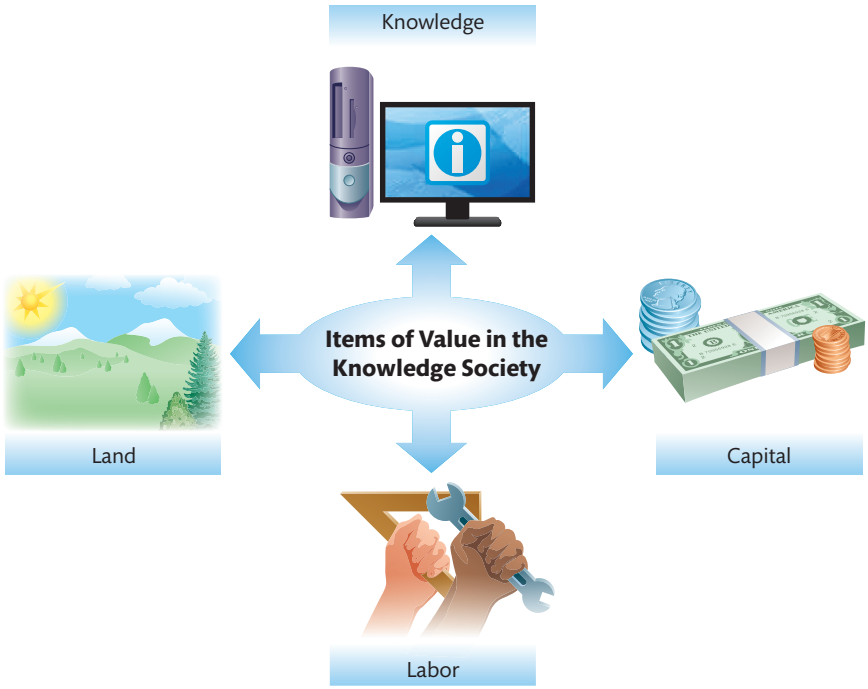
KNOWLEDGE WORKERS AND THE KNOWLEDGE SOCIETY. In 1959, Peter Drucker predicted that information and information systems would become increasingly important, and at that point, more than half a century ago, he coined the term **knowledge worker**. Knowledge workers are typically professionals who are relatively well educated and who create, modify, and/or synthesize knowledge as a fundamental part of their jobs.

Drucker's predictions about knowledge workers were accurate. As he predicted, they are generally paid better than their prior agricultural and industrial counterparts; they rely on and are empowered by formal education, yet they often also possess valuable real-world skills; they are continually learning how to do their jobs better; they have much better career opportunities and far more bargaining power than workers ever had before. Knowledge workers make up about a quarter of the workforce in the United States and in other developed nations, and their numbers are rising quickly.

Drucker also predicted that, with the growth in the number of knowledge workers and with their rise in importance and leadership, a **knowledge society** would emerge. He reasoned that, given the importance of education and learning to knowledge workers and the firms that need them, education would become the cornerstone of the knowledge society. Possessing knowledge, he argued, would be as important as possessing land, labor, or capital (if not more so) (Figure 1.3). Indeed, research shows that people equipped to prosper in the knowledge society, such as those with a college education, earn far more on average than people without a college education, and that gap is increasing. In fact, the most recent data from the U.S. Census Bureau's American Community Survey (2014 data) reinforce the value of a college education: Median earnings for workers 25 and over with a bachelor's degree were US$50,450 a year, while those for workers with a high school diploma were US$27,809. Median earnings for workers with a graduate or professional degree were US$66,175, and for those without a high school diploma US$20,542. These data suggest that a bachelor's degree is worth about US$1 million in additional lifetime earnings compared to a worker with only a high school diploma.

FIGURE 1.3

Knowledge has become as important as—and many feel more important than—land, labor, and capital resources.



Additionally, getting a college degree will qualify you for many jobs that would not be available to you otherwise and will distinguish you from other job candidates. Finally, a college degree is often a requirement to qualify for career advancement and promotion opportunities once you do get that job.

People generally agree that Drucker was accurate about knowledge workers and the evolution of society. While people have settled on Drucker's term *knowledge worker*, there are many alternatives to the term *knowledge society*. Others have referred to this phenomenon as the *knowledge economy*, the *new economy*, the *digital society*, the *network era*, the *Internet era*, and other names. We simply refer to this as the *digital world*. All these ideas have in common the premise that information and related technologies and systems have become indispensable and that knowledge workers are vital.

Today, not only knowledge workers use information systems as integral parts of their work lives; many "traditional" occupations now increasingly use information systems—from the UPS package delivery person using global positioning system (GPS) technology to take the best route to deliver parcels to the farmer in Iowa who uses precision agriculture to plan the use of fertilizers to increase crop yield. In essence, (almost) every organization can now be considered an e-business. An **e-business** is an organization that uses information technologies or systems to support nearly every part of its business. Thus, the lines between "knowledge workers" and "manual workers" are blurring, to the point that some argue that "every worker is a knowledge worker" (Rosen, 2011).

THE DIGITAL DIVIDE. Some have argued, however, that there is a downside to being a knowledge worker and to living in the digital world. For example, some have argued that knowledge workers will be the first to be replaced by automation with information systems. Others have argued that in the new economy there is a **digital divide**, where those with access to information systems have great advantages over those without access to information systems. The digital divide is one of the major ethical challenges facing society today when you consider the strong linkage between computer literacy and a person's ability to compete in the digital world. For example, access to raw materials and money fueled the Industrial Revolution, "but in the informational society, the fuel, the power, is knowledge," emphasized John Kenneth Galbraith, an American economist who specialized in emerging trends in the U.S. economy. "One has now come to see a new class structure divided by those who have information and those who must function out of ignorance. This new class has its power not from money, not from land, but from knowledge" (Galbraith, 1987).

The good news is that the digital divide in America is rapidly shrinking, but there are still major challenges to overcome. In particular, people in rural communities, the elderly, people with disabilities, and minorities lag behind national averages for Internet access and computer literacy. Outside the United States and other developed countries, the gap gets even wider and the obstacles get much more difficult to overcome, particularly in the developing countries where infrastructure and financial resources are lacking. For example, most developing countries are lacking modern informational resources such as affordable Internet access or efficient electronic payment methods.

To be sure, there is a downside to overreliance on information systems, but one thing is for certain: Knowledge workers and information systems are now critical to the success of modern organizations, economies, and societies. At the same time, information systems play a crucial role in various major issues societies face. These issues are examined next.

Globalization and Societal Issues in the Digital World

The past decades have brought about a number of dramatic global changes, many of which will continue to influence individuals, businesses, economies, and societies well into the future. Many of such interrelated societal "megatrends," discussed by consulting firms such as PricewaterhouseCoopers (PwC) or Ernst & Young (EY), local and national governments, or global political and business leaders at the World Economic Forum, are related to ever-increasing **globalization**—the integration of economies throughout the world, enabled by innovation and technological progress (International Monetary Fund, 2002). You can see the effects of globalization in many ways, such as the greater international movement of commodities, money, information, and labor as well as the development of technologies, standards, and processes to facilitate this movement.



COMING ATTRACTIONS

Memory Crystals

In the *Superman* films and many other sci-fi movies and books, characters make use of data storage devices that resemble large crystals. In the stories, these crystals often store incredibly large amounts of data and last for extraordinary lengths of time. Now scientists have taken a step toward making such technology a reality. Researchers at the University of Southampton (UK) have created a nanostructured glass storage device that resembles the fictional technologies. The technique uses self-assembling nanostructures written into fused quartz using tiny femtosecond (one-quadrillionth, or one-millionth of one-billionth, of a second) laser light pulses. The data are encoded in five dimensions (5D): height, length, width, position, and orientation. Using these multiple dimensions along with the nanoscale laser writing allows a small glass disc, about the size of a large coin, to store 360 terabytes (TB) of data. As a terabyte is equal to 1,024 gigabytes (GB), the amount of data stored on each tiny disk is several hundred times the amount of data stored on a standard desktop computer (1–4 TB) and several thousand times the data storage capacity of most smartphones (16–128 GB). The quartz material is highly stable (up to 13.8 billion years at 190 degrees Celsius), so data can be archived essentially forever.

To demonstrate the technology, the scientists recorded several major documents from human history on the disks, including the Universal Declaration of Human Rights, Newton's *Opticks*, the Magna Carta, and the King James Bible. The technology could be used by any organization or business seeking to store large volumes of data for long periods of time. Museums, libraries, national archives, and others could preserve their information and records for nearly unlimited time. Data stored using the technique could well outlast any other aspects of not just our technology but our civilization. Professor Peter Kazansky from the university's research team says: "It is thrilling to think that we have created the technology to preserve documents and information and store it in space for future generations. This technology can secure the last evidence of our civilization: all we've learnt will not be forgotten." The scientists are looking for industry partners to further develop and commercialize the technology.

Based on:

Mullen, M. (2016, February 17). New "Superman" crystals can store data for billions of years. *CNN Money*. Retrieved April 14, 2016, from http://money.cnn.com/2016/02/17/technology/5d-data-storage-memory-crystals/index.html

Phys.org. (2016, February 15). Eternal 5D data storage could record the history of humankind. Retrieved April 14, 2016, from http://phys.org/news/2016-02-eternal-5d-storage-history-humankind.html

GLOBALIZATION: OPPORTUNITIES AND CHALLENGES. For organizations, globalization has opened up many opportunities, brought about by falling transportation and telecommunication costs. Today, shipping a bottle of wine from Australia to Europe costs merely a few cents, and people can make voice or video calls around the globe for free using services such as Skype, Google Hangouts, or WhatsApp. To a large extent fueled by movies, television, and other forms of media, the increasing globalization has moved cultures closer together. The streaming movie provider Netflix is available in almost every country of the world, people in all corners of the world can receive television programming from other countries, and major movies are increasingly international. Developments such as these help create a shared understanding about norms of behavior or interaction, desirable goods or services, or even forms of government (though such shared understanding is still often lacking, and many of these developments have not yet happened). The rapid rise of a new middle class in many developing countries has enabled established companies to reach new markets, enabling them to sell their products to literally millions of new customers. At the same time, with the decrease in communication costs, companies can now draw on a large pool of skilled professionals from all over the globe. Countries such as Russia, China, and India offer high-quality education, leading to an ample supply of well-trained people at low cost. Some countries have even built entire industries around certain competencies, such as software development or tax preparation in India and call centers in Ireland.

The tremendous decrease in communication costs has increased the use of **outsourcing**—the moving of business processes or tasks (such as accounting, manufacturing, or security) to another company or another country—as now companies can outsource business processes on a global scale (Figure 1.4). Companies are choosing to outsource business activities for a variety of reasons; the most important reasons include the following (King, 2003):

- To reduce or control costs
- To free up internal resources
- To gain access to world-class capabilities
- To increase the revenue potential of the organization
- To reduce time to market
- To increase process efficiencies
- To be able to focus on core activities
- To compensate for a lack of specific capabilities or skills

Often, companies located in countries such as India can provide certain services much cheaper because of lower labor costs, or companies perform certain functions in a different country to reduce costs or harness skilled labor. For example, in India, two companies—Wipro and Infosys—have emerged as the leaders in providing IT services that range from business consulting to systems development. In addition, a wide variety of other services—ranging from telephone support to tax returns—are candidates for outsourcing to different countries, be it Ireland, China, or India. Even highly specialized services, such as reading of X-rays by skilled

FIGURE 1.4

Companies are outsourcing production to overseas countries (such as China) to utilize talented workers or reduce costs.
Source: Lianxun Zhang/Fotolia.



radiologists, are outsourced by U.S. hospitals to doctors around the globe, often while doctors in the United States are sleeping.

Yet globalization has also brought about a number of operational challenges for organizations. Organizations face governmental challenges related to differences in political systems, regulatory environments, laws, standards, or individual freedoms. Likewise, geoeconomic challenges include differences in infrastructure, demographics, welfare, or workers' expertise. Lastly, organizations face cultural challenges, such as dealing with differences in languages, beliefs, attitudes, religions, or life focus but also different viewpoints regarding intellectual property. As a result, companies intending to outsource services or production have to carefully choose outsourcing locations, considering numerous different factors, such as English proficiency, salaries, or geopolitical risk. While countries such as India remain popular, other formerly popular countries (such as Singapore, Canada, or Ireland) are declining because of rising salaries. With these shifts, outsourcers are constantly looking at nascent and emerging countries such as Bulgaria, Egypt, Ghana, Bangladesh, or Vietnam.

Obviously, organizations have to weigh the potential benefits (e.g., cost savings) and drawbacks (e.g., higher geopolitical risk or less experienced workers) of outsourcing to a particular country, and often, cost savings prove to be negligible due to added overhead, such as customs, shipping, or training as well as quality problems. In fact, *InformationWeek*, a leading publication targeting business IT users, found that 20 percent of the 500 most innovative companies in terms of using IT took back projects previously outsourced to another country. Nevertheless, IT outsourcing is big business: Research firm IDC forecasted the market for IT outsourcing to be $103 billion in 2019 (Tapper, 2015).

SOCIETAL ISSUES IN THE DIGITAL WORLD. The rapid development of transportation and telecommunication technologies, national and global infrastructures, and information systems as well as a host of other factors has created a number of pressing societal issues that will tremendously influence the world we live in (PWC, 2016; Schreiber, 2016). In this section, we will highlight a few of these issues (Figure 1.5). One such issue is **demographic changes**—changes in the structure of populations such as related to age, birth rates, and migration. While many countries in the developed world see rapidly aging populations, developing regions such as Africa are expected to rapidly rise in population, fueling a massive global population growth. These differences in demographic changes will also shift the balance of demand and supply of labor; further, differences in welfare are likely to further increase, and many countries are already experiencing both positive and negative effects of mass migrations. In addition, many regions of the world are seeing rapid **urbanization**—the movement of rural populations to urban areas, to a point where 50 percent of the world's population is now living in cities (PWC, 2016); sustaining this growth while providing livable environments for the inhabitants will pose major challenges. Another major trend is the global **shifts in economic power**—changes in countries' purchasing power and control over natural resources—where established economies are losing

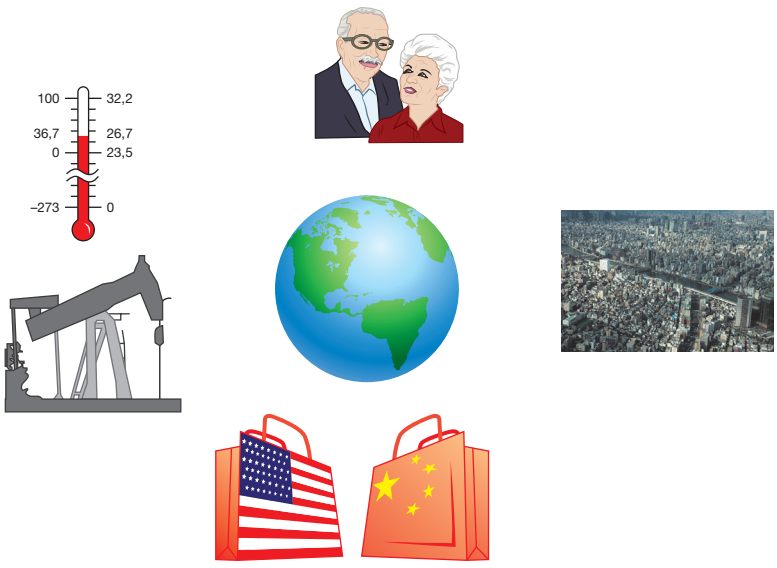


FIGURE 1.5
Societal issues in the digital world.
Source: Pichaitun/Fotolia.

their dominating positions in the world's economy, resulting in the need to resolve political struggles (PWC, 2016). Many of these issues interact, affect each other, and/or fuel other issues, such as issues related to **resource scarcity** due to limited availability of fossil fuels and other natural resources and **climate change**—large-scale and long-term regional and global changes in temperatures and weather patterns. Population growth, global trade, consumerism, and other factors contribute to increasing waste and pollution, as well as a growing need for resources, at a time where humans already live beyond the finite natural resources the planet can provide. Likewise, climate change—regardless of its causes—and its associated changes in weather patterns, rise in sea levels, and increase in the severity of storms pose a number of challenges for individuals, societies, and the world. As a consequence, **sustainable development**—"development that meets the needs of the present without compromising the ability of future generations to meet their own needs" (World Commission on Environment and Development, 1987)—will become an ever increasingly important aspect. In addition to these societal issues, we have witnessed a number of breakthroughs and transformations enabled by technology; these breakthroughs are disrupting traditional business models but can also help address pressing societal issues. Next we will discuss five IT megatrends that shape the digital future.

Five IT Megatrends That Shape the Digital Future

In most developed societies, information technologies have become pervasive—information technologies are in fact used throughout society, and the speed of innovations is increasing at a tremendous pace, with many *radical innovations* marginalizing or displacing existing products or industries (see Chapter 2, "Gaining Competitive Advantage Through Information Systems").



WHO'S GOING MOBILE

Wearable Technologies

For a long time, the smart fridge has been touted as the prime example of possibilities enabled by the *Internet of Things*, a network of physical objects that are connected over the Internet. While the smart fridge has yet to take off, many much smaller "things" have become tremendously popular—wearable technologies, such as smartwatches or activity trackers. The term **wearable technologies** refers to clothing or accessories that incorporate electronic technologies. For example, the Apple Watch, Samsung's Galaxy Gear, or the Fitbit incorporate various sensors; depending on the device, the sensors record physiological data such as body movements or heart rate but also environmental data such as ambient light, orientation, or altitude. Smartwatches such as the Apple Watch or Samsung's Galaxy Gear are designed to be an extension of the user's phones, used to display notifications from the phone or tablet devices, providing quick access to some of the phone's or tablet's functions, in addition to enabling the user to monitor various fitness activities. Activity trackers such as the Fitbit are designed to be worn and passively used on a regular basis, supporting the "**quantified self**"—the logging of all aspects of one's daily life, ranging from monitoring and recording of activities, performance, or intakes to monitoring bodily states (such as moods or physiological data) to improve one's overall health and performance. Yet other devices are intended for special use cases—as in the "Climbax," a rock-climbing device that tracks your climbing technique.

Whereas many wearable technologies are designed to support the quantified self, many see social relationships as the next big thing in wearables and experiment with social wearables that allow users to connect with each other in novel ways. For example, experimental concepts include a social fan jersey, which allows sports fans to experience vibration patterns as their favorite rugby player hits the ground. Many of these technologies are both exciting and futuristic, but there remain security, privacy, and other issues to resolve: Early wearable devices have been demonstrated to be easily hackable, and there is also broad concern about the potential distractions to automobile drivers or students in classrooms from such wearable devices. While offering great promises, these exciting technologies will require adjustments and compromises if they are to obtain the extensive adoption their manufacturers are aiming for.

Based on:

Cosco, A. (2016, March 7). The social age of wearable tech: From quantified self to emotional second skin. *Wareable*. Retrieved March 18, 2016, from http://www.wareable.com/wearable-tech/the-social-age-of-wearable-tech-beyond-the-quantified-self

Green, C. (2014, May 12). Wearable technology creates $50 billion investment frenzy. *Information Age*. Retrieved March 18, 2016, from http://www.information-age.com/technology/mobile-and-networking/123457988/wearable-technology-creates-50-billion-investment-frenzy

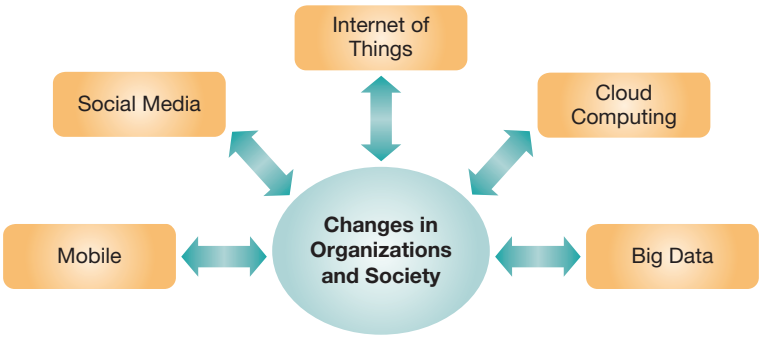


FIGURE 1.6
Five IT megatrends.

For example, within just a few years, drones evolved from being primarily used by the military to being used by farmers, aerial photographers, filmmakers, and hobbyists alike. Self-parking systems are already available in many vehicles, self-driving cars and trucks are being actively tested by various companies, and autonomous Caterpillar mining trucks are already in use. Likewise, the development of sophisticated web technologies has brought about a fundamental shift in types of information technologies that are being used, and we're seeing five (intertwined) "megatrends" that influence individuals, organizations, and society (Figure 1.6). Understanding the influence of these megatrends will be increasingly important: Individuals will increasingly feel the impact of these megatrends on their private and work lives, and businesses need to have a business strategy that is fit for today's digital world and the digital future.

- ***Mobile.*** One of the biggest trends we're seeing today is the move toward mobile devices, as indicated in the opening section of this chapter. In most developed countries, the vast majority of adults have a mobile phone, and typically, people have their mobile phones within their reach 24/7. Compare that with the access to your laptop or PC. In the developing world, mobile devices are frequently seen leapfrogging traditional PCs; owing to the lack of stable, reliable power or landline telephone infrastructure, mobile devices are often the primary means of accessing the Internet. For organizations, this increase in mobility has a wide range of implications, from increased collaboration to the ability to manage a business in real time—at any time, from anywhere—to changes in the way new (or existing) customers can be reached (Figure 1.7). With the increase in mobile devices, organizations not only have to create mobile-device-friendly versions of their websites but often

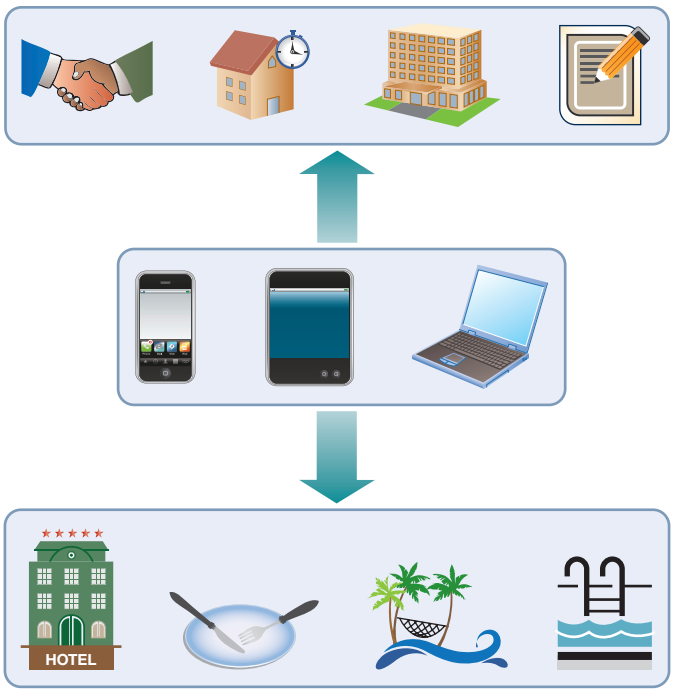


FIGURE 1.7
Mobile devices allow running business in real time—at any time, from anywhere.

build mobile **apps** (software programs designed to perform a particular, well-defined function) to market their products or services. In addition, fueled by advances in consumer-oriented mobile devices (such as smartphones and tablets) and the ability to access data and applications "in the cloud," today's employees are increasingly using their own devices for work-related purposes or are using software they are used to (such as social networks for communicating) in the workplace. While initially, workers tended to use their own devices primarily for checking e-mails or visiting social networking sites, they now use their own devices for various other important tasks, including customer relationship management or enterprise resource planning. For organizations, this trend can be worrying (due to concerns related to security or compliance or increasing need to support the workers' own devices), but it can also provide a host of opportunities, such as increased productivity, higher retention rates of talented employees, or higher customer satisfaction. Managing this trend of "bring your own device" (**BYOD**) is clearly a major concern of business and IT managers alike. Further, we have witnessed the **consumerization of IT**; many technological innovations are first introduced in the consumer marketplace before being used by organizations, and businesses have to constantly evaluate how a wide variety of new technologies might influence their ways of doing business. Throughout the text, we will introduce issues and new developments associated with increases in mobility.

- ***Social Media.*** A second megatrend, as you have undoubtedly noticed, is social media. The rise of social media is largely based on the **network effect**—referring to the notion that the value of a network (or tool or application based on a network) increases with the number of other users. In other words, if a network has few users, it has little or no value. You may be one of the more than 1.6 billion (and growing) Facebook users who share status updates or pictures with friends and family, or you may use apps such as Snapchat, Twitter, or WhatsApp to communicate with your friends. How useful would these social media platforms be if none of your friends or family members had access to them? Today, the use of social media has gone way beyond friends and families. University professors use social networks to provide students with updates about course-related topics, and organizations use social media to encourage employee collaboration or to connect with their customers (Figure 1.8). In addition, companies can harness the power of the crowd by using social media to get people to participate in innovation and other activities. With the rise of social media, we have witnessed a shift in roles, where users are not mere consumers of information but have become important creators and contributors. As you can imagine, social media are here to stay; while we will touch on social media–related aspects throughout the book, we will devote Chapter 5, "Enhancing Organizational Communication and Collaboration Using Social Media," to social media and related topics.
- ***The Internet of Things.*** A third megatrend is the **Internet of Things (IoT)**—a network of a broad range of physical objects that can automatically share data over the Internet. Such objects (or "things") can range from an automobile tire equipped with a pressure sensor to a smart meter enabling remote monitoring of energy consumption to a cow with an injectable ID chip. Already in 2008, more devices were connected to the Internet than there were people living on earth. Fueled by advances in chips and wireless radios and decreasing costs of **sensors** (devices that can detect, record, and report changes in the physical environment), in the not-too-distant future everything that can generate useful information



FIGURE 1.8
Social media are used in various personal and business settings.

will be equipped with sensors and wireless radios (Figure 1.9). In other words, anything that can generate data or uses data can be connected, accessed, or controlled via the Internet (sometimes referred to as "pervasive computing"). With the ability to connect "things" such as sensors, meters, signals, motors, actuators, or cameras, the potential for gathering useful data is almost limitless. For example, the market for **smart home technologies** (sometimes called **home automation**)—technologies enabling the remote monitoring and controlling of lighting, heating, or home appliances—such as the Nest Learning Thermostat, is expected to reach almost US$60 billion by 2020. Using smart home technology, one can monitor home temperatures when on vacation or remotely adjust the air-conditioning; likewise, sensors integrated in a road's surface can monitor temperatures and trigger dynamic speed limits in case there is the risk of ice or snow. Similarly, sensors can monitor availability of parking spaces or traffic flow, alerting drivers of changes in conditions. Millions of sensors connected to the Internet can monitor weather conditions, helping to generate more accurate local weather predictions, or can monitor soil moisture in golf courses, reducing the need for watering. Cardiac monitors can alert physicians of patients' health risks. The use of IoT technologies in manufacturing—referred to as **Industrial Internet of Things (IIoT)**—enables the convergence of information technology and operations technology, offering the potential for tremendous improvements in efficiency, product quality, agility, and flexibility, allowing companies to mass-produce customized products, better monitor supply chains, and so on. In sum, the applications of sensor technology for home automation, smart cities, smart metering, smart farming, e-health, manufacturing, and other areas are almost limitless. As the number of sensors and devices connected to the Internet grows, the Internet of Things will evolve to become the Internet of Everything (IoE), where just about any device's functionality is enhanced through connectivity and intelligence.

- ***Cloud Computing.*** The fourth megatrend is **cloud computing**. Whereas traditionally each user would install a number of different applications for various tasks—from creating documents to listening to music—as well as store documents, pictures, and other data on

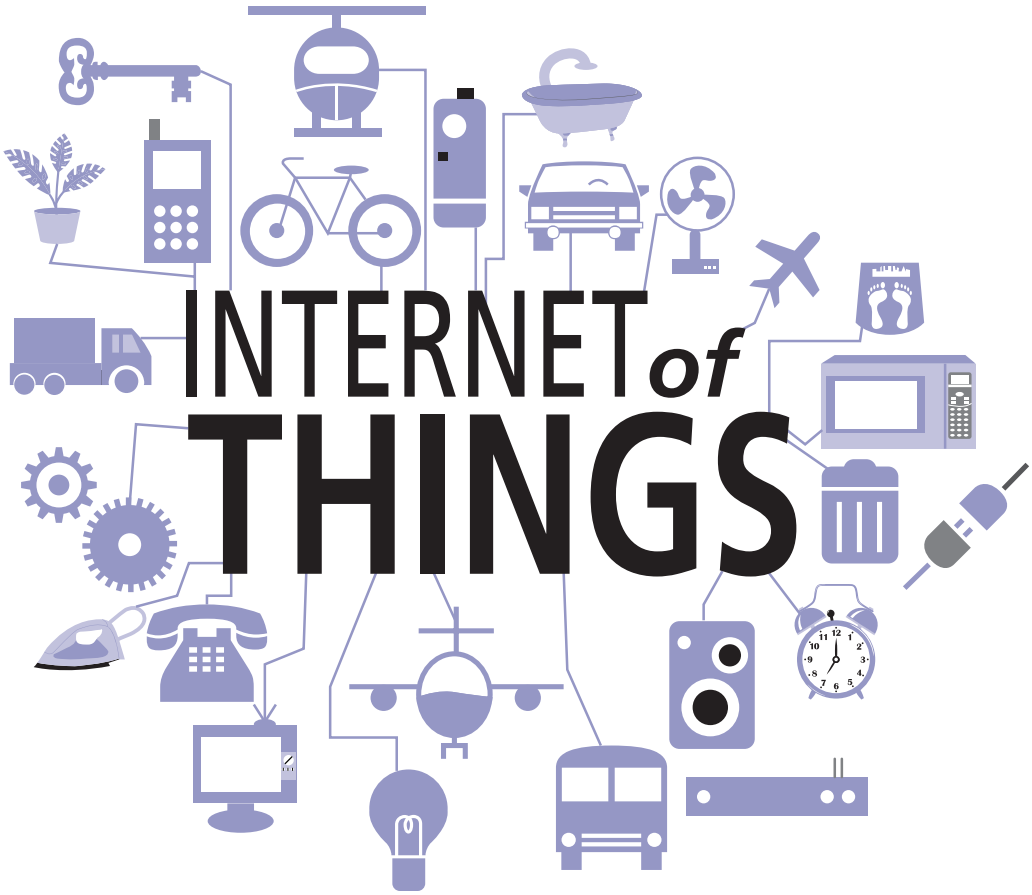


FIGURE 1.9
The Internet of Things.

FIGURE 1.10

Applications and data stored in the cloud can be accessed from different devices.



his or her computer, web technologies enable using the Internet as the platform for applications and data. Now, much of the functionality previously offered by applications installed on each individual computer is offered by applications "in the cloud," accessed via a web browser. Increasingly, not only the applications but also the data reside in the cloud, to be accessed at any time from anywhere (Figure 1.10). A good example of cloud computing is the various services offered by Google, such as Gmail (e-mail), Google Docs (word processing), and Google Calendar, all of which are accessed via a web browser, freeing users from the task of installing or updating traditional desktop applications or worrying about storing or backing up data. Cloud computing has made inroads in a variety of organizational applications, and many organizations rely on an information systems infrastructure in the cloud. Further, cloud computing can enable advanced analytics of massive amounts of Big Data generated by mobile devices, sensors, or users of social networks. We will extensively discuss cloud computing in Chapter 3, "Managing the Information Systems Infrastructure and Services."

- ***Big Data.*** Together, these transformations of our social and work interactions enabled by 24/7 connectivity have given rise to a fifth trend, **Big Data**. Big Data are typically described as extremely large and complex datasets, which are characterized as being of high volume, variety (i.e., many different types of data), and velocity (i.e., the data are being collected and analyzed at ever-increasing rates). Following the old adage that information is power, organizations are continuously seeking to get the right information to make the best business decisions. Yet organizations are generating and collecting ever more data from internal and external sources. The rise of social media has further increased the amount of unstructured data available to organizations; for example, people frequently voice their thoughts about products or companies on blogs or social networks. In addition, the Internet of Things, allowing for connecting devices and sensors to the Internet, further contributes to the growth of data available to organizations and individuals. With decreasing costs for capturing and storing data, data are now not only ubiquitous but also cheap. A study by research firm IDC estimated that in 2013, 4.4 zettabytes of data were generated and consumed. How much is 4.4 zettabytes? Well, 4.4 zettabytes equals 4.4 trillion gigabytes, or the equivalent of 32 billion 128GB iPads. As the number of devices connected to the Internet is expected to reach 30 billion in 2020, the amount of digital data generated worldwide is forecast to reach 44 zettabytes in 2020 and 180 zettabytes by 2025. For many organizations today, value is created from data. Consider, for example, that the largest/most valuable organizations in the "old economy" (such as GE, Dow, or Ford) have 100,000–300,000 employees, and the largest organizations in the "new economy" (such as Microsoft, HP, or Oracle) have 50,000–100,000 employees; in contrast, modern companies of the digital world (such as Facebook, Twitter, or Groupon) have risen to the top with a mere 5,000–15,000 employees by creating value from data (Hofmann, 2011) (Figure 1.11). Ever-increasing amounts of data increase the ability to detect meaningful relationships and regularities, and insights gained from analyzing Big Data not only can contribute to business success but can also help to address some of the tremendous challenges society faces.

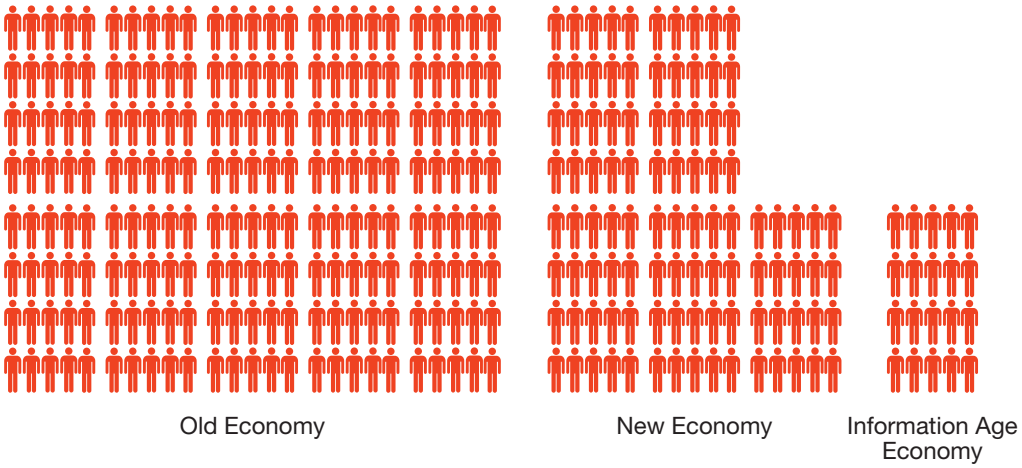


FIGURE 1.11

Companies in the Information Age economy are creating value not from people but from data.

For example, Big Data is a key factor enabling research ranging from genomics to climate change. However, analyzing tremendous amounts of (often unstructured) data (i.e., Big Data) poses tremendous challenges for organizations. In Chapter 6, "Enhancing Business Intelligence Using Big Data and Analytics," we will discuss how organizations can harness Big Data to make better business decisions.

Just as the societal issues interact and affect each other, so do the IT megatrends. For example, cloud computing allows for generating new machine learning algorithms to analyze Big Data, continuously improving *artificial intelligence* capabilities. Similarly, continuous input from various sensors, paired with artificial intelligence to make sense of such Big Data streams, enables tremendous advances in **robotics** (i.e., the use of robots to perform manual tasks). The different megatrends have also enabled various business model innovations that disrupt established industries. For example, fueled by mobile devices, social media, and cloud computing, Uber and Airbnb are examples of innovative business models based on the concept of a *sharing economy* and disrupt traditional taxi and lodging companies. Likewise, the Internet of Things and the massive amounts of data generated enable the creation of service-oriented business models (sometimes referred to *servitization*), where companies shift from selling physical products to providing these as services (see Chapter 2); for example, using sensors to monitor performance, temperature, or mileage enables tire manufacturers Bridgestone and Michelin to sell tires as a service, where truck operators pay based on usage, whereas the manufacturer is responsible for the tires' performance. Other industries being disrupted range from the financial industry to healthcare providers, where information systems allow for various radical innovations.

While these megatrends open up an almost unlimited potential for innovative products, services, or processes, they also pose a variety of challenges for organizations operating in the digital world. Throughout the book, we will discuss not only the opportunities but also the challenges organizations face when trying to harness the potential of these megatrends. What do these megatrends mean for you and for today's workforce? On a most basic level, they imply that being able to use information systems, to assess the impacts of new technologies on one's work or private life, and to learn new technologies as they come along will be increasingly important skills.

Most modern-day high school and university students have grown up in a computerized world. If by some chance they do not know how to operate a computer by the time they graduate from high school, they soon acquire computer skills because in today's work world, knowing how to use a computer—called **computer literacy** (or information literacy)—can not only open up myriad sources of information but can also mean the difference between being employed and being unemployed. In fact, some fear that the Information Age will not provide the same advantages to "information haves"—those computer-literate individuals who have almost unlimited access to information—and "information have-nots"—those with limited or no computer access or skills.

Computer-related occupations have evolved as computers have become more sophisticated and more widely used. Where once we thought of computer workers as primarily programmers, data entry clerks, systems analysts, or computer repairpersons, today many more job categories in virtually all industries, from accounting to the medical field, involve the use of computers. In



GREEN IT

The Green Internet of Things

The Internet and associated technologies have been busy disrupting business and society for the past several decades. Next up, another revolution in information technology is going to shake things up again. Green IT (or green computing, see Chapter 3) refers to the study and practice of using computing resources more efficiently to reduce environmental impacts as well as the use of information systems to reduce negative environmental impacts. The Internet of Things brings connectivity and information technology to places never before considered. Together, these technologies are once again poised to revolutionize business and society.

Traditionally, IT resources were seen as an ever-expanding pool—as business needs grew, more servers and data centers were installed. Eventually a limit has to be reached; the impact of power consumption alone from a modern data center can be profound. New technologies and techniques are having a large impact on both how we provision IT resources and how we interact with our world's resources. New servers are designed for low power consumption. Cloud computing architectures allow resources to be allocated on an as-needed basis.

With the Internet maturing into an established platform, new opportunities have become apparent. By combining ubiquitous connectivity with inexpensive processing power and sensor devices, nearly anything can be connected to the Internet. To be considered a part of the IoT, a device simply needs to be connected to the Internet, collect and transmit sensory data, and be something physical that interacts with the real world. Umbrellas notify us of the weather, smartwatches monitor our steps and vital signs, and nano-scale sensors are helping scientists collect unprecedented data about natural phenomena and ecosystems. Power companies can instrument our houses, our cars, and their distribution systems to gain unprecedented insights into energy use and demand.

Internet technologies disrupted many businesses and social processes by changing the scope and scale of interactions between people. By making large-scale interaction and communication possible almost instantaneously, supply chains could be redesigned, globalization was accelerated, and political processes were altered (for better and for worse). Individuals became citizen journalists. Together with the IoT, green technologies are enabling more accurate forecasting of resource needs and allow businesses and governments alike to become more informed and responsive. Tomorrow's leaders will need to incorporate such devices and systems into their planning to stay ahead of customer and citizen wants and needs.

Based on:

Byles, D. (2016, January 13). Technology, disruption, the Internet of Things… and you. *Businessgreen.com*. Retrieved April 24, 2016, from http://www.businessgreen.com/bg/opinion/2441499/technology-disruption-the-internet-of-things-and-you

McCabe, L. (2009, December 30). What is green IT, and why should you care? *Smallbusinesscomputing.com*. Retrieved April 24, 2016, from http://www.smallbusinesscomputing.com/testdrive/article.php/3855806/What-Is-Green-IT-and-Why-Should-You-Care.htm

fact, today there are few occupations where computers are not somehow in use. Computers manage air traffic, perform medical tests, monitor investment portfolios, control construction machinery, and more. Because they are especially adept at processing large amounts of data, they are used extensively by universities and public schools, in businesses of all sizes, and in all levels and departments of government. Engineers, architects, interior designers, and artists use special-purpose computer-aided design programs. Musicians play computerized instruments, and they write and record songs with the help of computers. Professionals in the medical industry use **healthcare IS** to support everything from patient diagnosis and treatment to analyzing patient and disease data to running doctors' offices and hospitals (see Chapter 6). Not only do we use computers at work, we also use them in our personal lives. We teach our children on them, manage our finances, do our taxes, compose letters and term papers, create greeting cards, send and receive e-mail, surf the Internet, purchase products, and play games on them. With the increasing use of computers in all areas of society, many argue that being computer literate—knowing how to use a computer and use certain applications—is not sufficient in today's world; rather, **computer fluency**—the ability to independently learn new technologies as they emerge and assess their impact on one's work and life—is what will set you apart in the future.

Information Systems Defined

An **information system (IS)** is the combination of people and information technology that create, collect, process, store, and distribute useful data. **Information technology (IT)** includes **hardware**, **software**, and **telecommunications networks**. Hardware refers to physical computer

equipment, such as a computer, tablet, or printer, as well as components like a computer monitor or keyboard. Software refers to a program or set of programs that tell the computer to perform certain tasks. Telecommunications networks refer to a group of two or more computer systems linked together with communications equipment. Although we discuss the design, implementation, use, and implications of hardware, software, and telecommunications throughout the text, the specifics on hardware, software, and telecommunications networks are discussed in Chapter 3 and the Technology Briefing. While traditionally the term *information technology* referred to the hardware, software, and networking components of an information system, the difference is shrinking, with many using the terms *IS* and *IT* synonymously. It is important to note that while many of today's technologies operate autonomously, they don't build themselves and do not exist for their own sake; rather, they are created to serve a useful purpose for people. Also, any information system involves data that are useful, for someone, somewhere. For example, transactional data are useful for businesses, status updates in your news feed on Facebook are useful for your friends as well as for Facebook itself, scores in a computer game are useful for the player as well as for the game developers, and so on. In Figure 1.12, we show the relationships among these IS components.

People in organizations use information systems to process sales transactions, manage loan applications, or help financial analysts decide where, when, and how to invest. Product managers also use them to help decide where, when, and how to market their products and related services, and production managers use them to help decide when and how to manufacture products. Information systems also enable us to get cash from ATMs, communicate by live video with people in other parts of the world, or buy concert or airline tickets. (Note that the term *information systems* is also used to describe the field comprising people who develop, use, manage, and study information systems in organizations.)

It is important to note that people use various terms to describe the field of information systems, such as *management information systems*, *business information systems*, *computer information systems*, and simply *systems*. Next, we more thoroughly examine the key components of the IS definition.

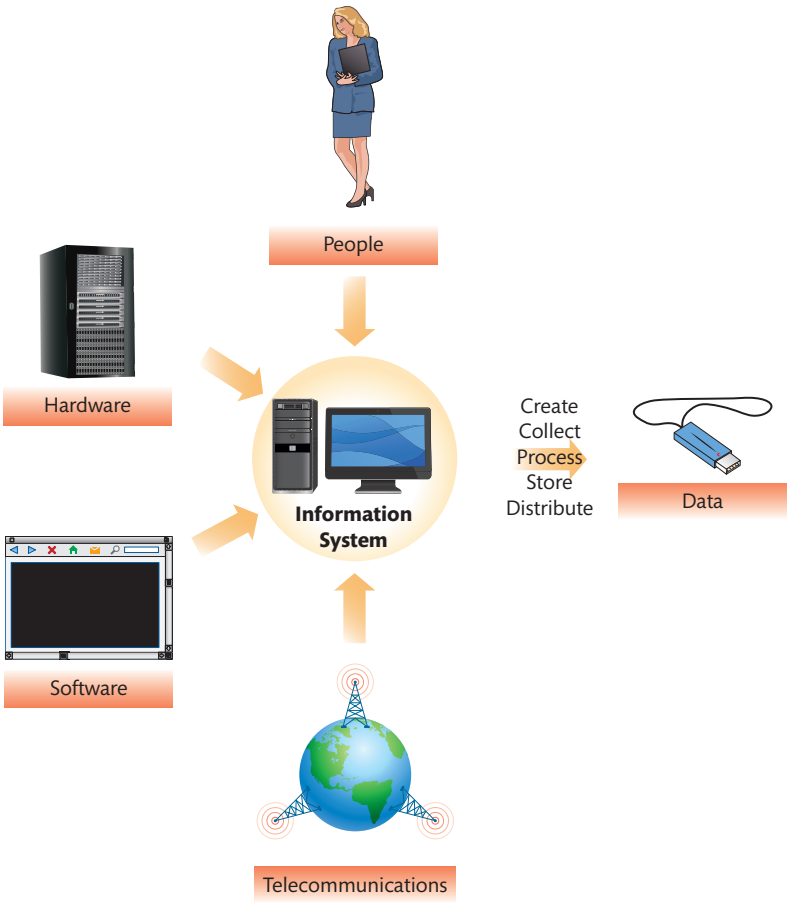


FIGURE 1.12

An information system is the combination of people and information technology that create, collect, process, store, and distribute useful data.

Data: The Root and Purpose of Information Systems

Earlier, we defined information systems as the combination of people and information technology that create, collect, process, store, and distribute useful data. We begin by talking about data, the most basic element of any information system.

DATA. Before you can understand how information systems work, it is important to distinguish between raw, unformatted data, information, and knowledge. Unformatted data, or simply **data**, are raw symbols, such as characters and numbers. Data have no meaning in and of themselves and are of little value until processed (Ackoff, 1989). For example, if we asked you what 465889727 meant or stood for, you could not tell us (Figure 1.13). However, if we presented the same data as 465-88-9727 and told you it was located in a certain database, in John Doe's record, in a field labeled *SSN*, you might rightly surmise that the number was actually the Social Security number of someone named John Doe. While data have no inherent meaning, the old adage "garbage in, garbage out" applies to data as well; thus, a key consideration of assessing whether data are reliable for making decisions is **data quality**, consisting of completeness, accuracy, timeliness, validity, and consistency.

INFORMATION. Data can be formatted, organized, or processed to be *useful*; they are transformed into **information**, which can be defined as a representation of reality, and can help to answer questions about who, what, where, and when (Ackoff, 1989). In the previous example, 465-88-9727 was used to represent and identify an individual person, John Doe (see Figure 1.13). Contextual cues, such as a label, are needed to turn data into information that is familiar and useful to the reader. Think about your experience with ATMs. A list of all the transactions at a bank's ATMs over the course of a month would be fairly useless data. However, a table that divided ATM users into two categories, bank customers and non-bank customers, and compared the two groups' use of the machine—their purpose for using the ATMs and the times and days on which they use them—would be incredibly useful information. A bank manager could use this information to create marketing mailings to attract new customers. Without information systems, it would be difficult to transform raw data into useful information.

KNOWLEDGE. In order to actually use information, knowledge is needed. **Knowledge** is the ability to understand information, form opinions, and make decisions or predictions based on the information. For example, you must have knowledge to be aware that only one Social Security number can uniquely identify each individual (see Figure 1.13). Knowledge is a body of governing procedures, such as guidelines or rules, that are used to organize or manipulate data to make them suitable for a given task.

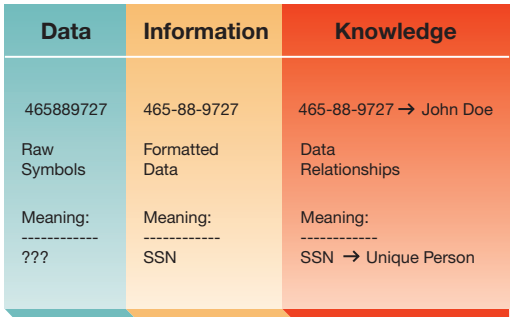
Understanding the distinctions between data, information, and knowledge is important because all are used in the study, development, and use of information systems.

Hardware, Software, and Telecommunications Networks: The Components of Information Systems

Ever since the dawn of humankind, there was a need to transform data into useful information for people, and people have invented various calculating devices, such as the abacus or the slide rule. Before the introduction of the first computers (which worked on a mechanical basis using punch cards), almost all business and government information systems consisted of file folders,

FIGURE 1.13

Data, information, and knowledge.



filing cabinets, and document repositories. Computer hardware has replaced these physical artifacts, providing the technologies to input and process data and output useful information; today, hardware includes not only "traditional" computer components but a variety of other input and output devices, including sensors, cameras, actuators, and the like. Software enables organizations to utilize the hardware to execute their business processes and competitive strategy by providing the computer hardware with instructions on what processing functions to perform. Finally, the telecommunications networks allow computers to share data and services, enabling the global collaboration, communication, and commerce we see today. The rapid evolution of the various hardware, software, and networking components make the ability to tie everything together ever more important.

People: The Builders, Managers, and Users of Information Systems

The IS field includes a vast collection of people who develop, maintain, manage, and study information systems. Yet an information system does not exist in a vacuum and is of little use if it weren't for you—the user. We will begin by discussing the IS profession and then talk about why knowing about fundamental concepts of information systems is of crucial importance in your personal and professional life.

If you are choosing a career in the IS field, you will find countless opportunities. With the growing value of data for competitive advantage, every company can now be considered a technology company, needing people with the right skill set to help optimize its business processes. The career opportunities for a person with IS training continue to be strong, and they are expected to continue to improve over the next 10 years. For example, the 2016–17 edition of the *Occupational Outlook Handbook* published by the U.S. Bureau of Labor Statistics predicted that employment for computer and IS managers will grow 15 percent through 2024, much faster than the average for all occupations (www.bls.gov/ooh/management/computer-and-information-systems-managers.htm). As more and more organizations rely more heavily on IS professionals, this boost in employment will occur in nearly every industry, not just computer hardware and software companies. *Money* magazine (http://money.cnn.com/pf/best-jobs) ranked software architect as the best job in America, with database developer and information assurance analyst also being among the top 10 best jobs in America (Table 1.1); also, *U.S. News* magazine (http://money.usnews.com/careers/best-jobs/rankings/the-100-best-jobs) rated computer systems analyst and web developer among the top 20 jobs (two of the few nonmedical jobs in that list). Likewise, a degree in information systems can provide the foundation for becoming a data scientist, currently one of the jobs with highest demand (Florentine, 2016).

In addition to an ample supply of jobs, earnings for IS professionals will remain strong. According to the U.S. Bureau of Labor Statistics, median annual earnings of these managers in May 2014 were US$127,640, with the top 10 percent earning more than US$187,200. Also, according to Salary.com, the median salary in 2016 for IT managers was US$109,701. According to a 2016 report by the National Association of Colleges and Employers, management

TABLE 1.1 Best Jobs in America (2015)

Rank	Career	Job Growth (10-year forecast)	Median Pay (in US$)
1	Software architect	23%	124,000
2	Video game designer	19%	79,900
3	Landman	13%	103,000
4	Patent agent	13%	126,000
5	Hospital administrator	23%	114,000
6	Continuous improvement manager	12%	96,600
7	Clinical nurse specialist	19%	89,300
8	Database developer	23%	88,200
9	Information assurance analyst	37%	96,400
10	Yoga instructor	13%	62,400

Source: Based on 100 Best Jobs in America, published by CNN Money, © 2016.

information systems was expected to be the highest-paid business major, with a mean starting salary of US$56,846. Likewise, information systems graduates with a master's degree had an average starting salary of US$67,632, higher than business majors such as accounting, finance, or marketing, according to a study by Temple University. Finally, computer and IS managers, especially those at higher levels, often receive more employment-related benefits—such as expense accounts, stock option plans, and bonuses—than do nonmanagerial workers in their organizations (a study by Payscale.com found that IS majors were—post-graduation—among the most satisfied with their careers).

As you can see, there continues to be a very strong need for people with IS knowledge, skills, and abilities—in particular, people with advanced IS skills, as we describe here. In fact, IS careers are regularly selected as not only one of the fastest growing but also a career with far-above-average opportunities for greater personal growth, stability, and advancement. Although technology continues to become easier to use, there is still and is likely to continue to be an acute need for people within the organization who have the responsibility of planning for, designing, developing, maintaining, and managing technologies. Much of this will happen within the business units and will be done by those with primarily business duties and tasks as opposed to systems duties and tasks. However, we are a long way from the day when technology is so easy to deploy that a need no longer exists for people with advanced IS knowledge and skills. In fact, many people believe that this day may never come. Although increasing numbers of people will incorporate systems responsibilities within their nonsystems jobs, there will continue to be a need for people with primarily systems responsibilities. In short, IS staffs and departments will likely continue to exist and play an important role in the foreseeable future.

Given that information systems continue to be a critical tool for business success, it is not likely that IS departments will go away or even shrink significantly. Indeed, all projections are for long-term growth of information systems in both scale and scope. Also, as is the case in any area of business, those people who are continually learning, continuing to grow, and continuing to find new ways to add value and who have advanced and/or unique skills will always be sought after, whether in information systems or in any area of the firm.

The future opportunities in the IS field are likely to be found in a variety of areas, which is good news for everyone. Diversity in the technology area can embrace us all. It really does not matter much which area of information systems you choose to pursue—there will likely be a promising future there for you. Even if your career interests are outside information systems, being a well-informed and capable user of information technologies will greatly enhance your career prospects.

CAREERS IN INFORMATION SYSTEMS. The field of information systems includes those people in organizations who design and build systems, those who use these systems, and those responsible for managing these systems. The people who help develop and manage systems in organizations include systems analysts, systems programmers, systems operators, network administrators, database administrators, systems designers, systems managers, and chief information officers. (In Table 1.2 we describe some of these careers.) This list is not exhaustive; rather, it is intended to provide a sampling of IS management positions. Furthermore, many firms will use the same job title, but each is likely to define it in a different way, or different companies will have different titles for the same basic function. As you can see from Table 1.2, the range of career opportunities for IS managers is broad, and salary expectations are high.

WHAT MAKES IS PERSONNEL SO VALUABLE? In addition to the growing importance of people in the IS field, there have been changes in the nature of this type of work. No longer are IS departments in organizations filled only with nerdy men with pocket protectors. Many more women are in IS positions now. Also, it is now more common for an IS professional to be a polished, professional businessperson who can speak fluently about both business and technology. IS personnel are now well-trained, highly skilled, valuable professionals who garner high wages and prestige and who play a pivotal role in helping firms be successful.

Many studies have been aimed at helping us understand what knowledge and skills are necessary for a person in the IS area to be successful. Interestingly, these studies also point out just what it is about IS personnel that makes them so valuable to their organizations. In a nutshell, good IS personnel possess valuable, integrated knowledge and skills in three areas—technical, business, and systems—as outlined in Table 1.3 (see also Figure 1.14).

TABLE 1.2 Some IS Management Job Titles and Brief Job Descriptions

IS Activity	Job Title	Job Description	Salary Range, in US$, in Percentiles (25%–75%)
Develop	Systems analyst	Analyze business requirements and select information systems that meet those needs	63,000–80,000
	Software developer	Code, test, debug, and install programs	71,000–91,000
	Software architect	Create customized software for large corporations	98,000–130,000
	IT consultant	Provide IT knowledge to external clients	57,000–93,000
	Database developer	Develop, modernize, and streamline databases	57,000–68,000
Maintain	IT auditor	Audit information systems and operating procedures for compliance with internal and external standards	64,000–83,000
	Database administrator	Manage database and database management software use	75,000–99,000
	Webmaster	Manage a firm's website	59,000–80,000
Manage	IT manager	Manage existing information systems	95,000–126,000
	IS security manager	Manage security measures and disaster recovery	102,000–125,000
	Information assurance analyst	Ensure availability and security of information stored on networks and in the cloud	60,000–93,000
	E-commerce manager	Manage development, maintenance, and strategy related to e-commerce systems	97,000–126,000
	Chief information officer (CIO)	Highest-ranking IS manager; oversee strategic planning and IS use throughout the firm	213,000–316,000
	Chief digital officer (CDO)	Executive focused on converting traditional "analog" businesses to digital; oversee operations in rapidly changing digital sectors like mobile apps and social media	150,000–200,000
Study	University professor	Teach undergraduate and graduate students; study the use of information systems in organizations and society	70,000–180,000
	Government scientist	Perform research and development of information systems for homeland security, intelligence, and other related applications	60,000–200,000

Source: Based on http://www.salary.com, http://www.payscale.com.

TABLE 1.3 IS Professional Core Competencies

Domain	Description
Technical Knowledge and Skills	
Hardware	Hardware platforms, infrastructure, cloud computing, virtualization, peripherals, mobile devices
Software	Operating systems, application software, mobile apps
Networking	Network administration, cabling and network interface cards, wireless, Internet, security
Business Knowledge and Skills	
Business integration, industry	Business processes, functional areas of businesses and their integration, industry characteristics
Managing people and projects	Planning, organizing, leading, controlling, managing people and projects
Social	Interpersonal, group dynamics, political
Communication	Verbal, written, and technological communication and presentation
Systems Knowledge and Skills	
Systems integration	Connectivity, compatibility, integrating subsystems and systems
Development methodologies	Steps in systems analysis and design, systems development life cycle, alternative development methodologies
Critical thinking	Challenging one's and others' assumptions and ideas
Problem solving	Information gathering and synthesis, problem identification, solution formulation, comparison, choice

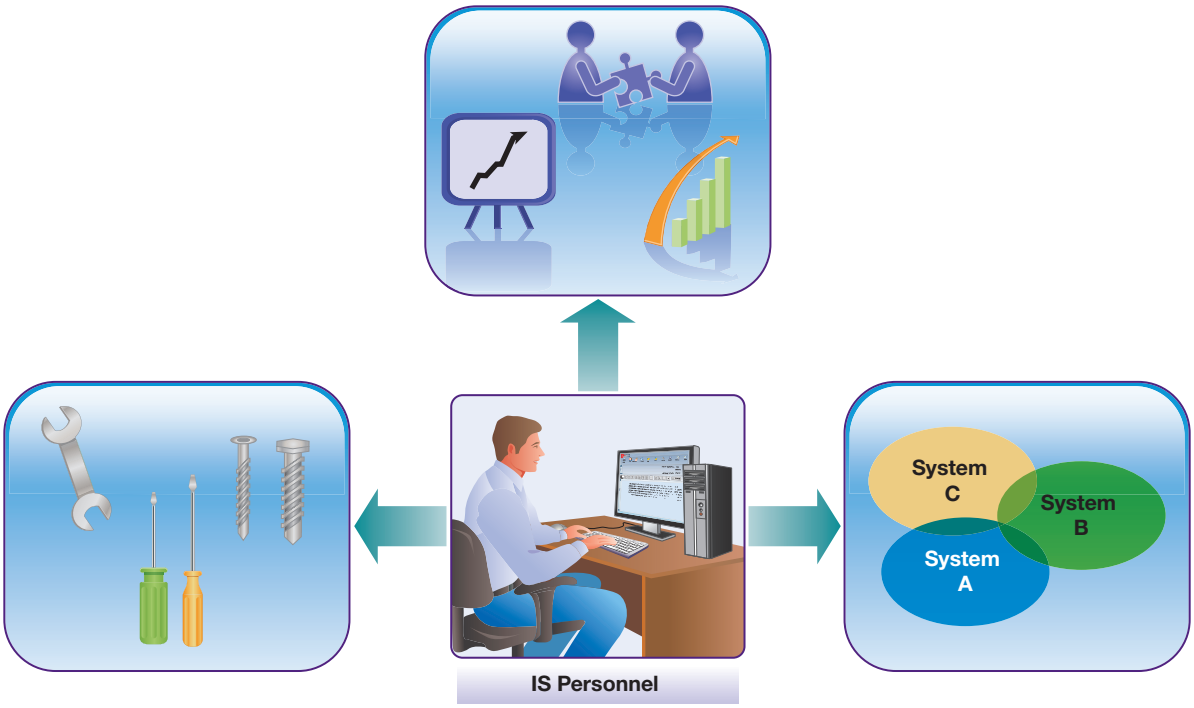


FIGURE 1.14

Good IS personnel possess valuable, integrated knowledge and skills in three areas—technical, business, and systems.

Technical Competency The technical competency area includes knowledge and skills in hardware, software, networking, and security. In a sense, this is the "nuts and bolts" of information systems. This is not to say that the IS professional must be a technical expert in these areas. On the contrary, the IS professional must know just enough about these areas to understand how they work, what they can do for an organization, and how they can and should be applied. Typically, the IS professional manages or directs those who have deeper, more detailed technical knowledge.

The technical area of competency is, perhaps, the most difficult to maintain because of the rapid pace of technological innovation in the digital world. With the economy rebounding, organizations are starting new projects or are reviving projects put on hold during the economic downturn; hence, while it once appeared as if most programming jobs or support jobs would be outsourced to third-party providers abroad, there is an increased demand in many companies for people with application development skills, especially in combination with sound business analysis and project management skills (Brandel, 2013). In fact, many of the hot skills listed in Table 1.4 are focused on the business domain, which is discussed next.

Business Competency The business competency area is one that sets the IS professional apart from others who have only technical knowledge and skills, and in an era of increased outsourcing, it may well save a person's job. For example, even though some low-level technology jobs may be outsourced, the Bureau of Labor Statistics recently reported that there is an increased need for IS managers as organizations embrace mobility and cloud computing (www.bls.gov/ooh/management/computer-and-information-systems-managers.htm). As a result, it is absolutely vital for IS professionals to understand the technical areas *and* the nature of the business. IS professionals must also be able to understand and manage people and projects, not just the technology. These business skills propel IS professionals into project management and, ultimately, high-paying middle- and upper-level management positions.

Systems Competency Systems competency is another area that sets the IS professional apart from others with only technical knowledge and skills. Those who understand how to build and integrate systems and how to solve problems will ultimately manage large, complex systems projects as well as manage those in the firm who have only technical knowledge and skills.

TABLE 1.4 Hot Skills for the Next Decade

Domain	Hot Skills
Business	Business–IT alignment; business analysis; enterprise solutions; business process modeling; project management; third-party provider management; enterprise-oriented social media
Technology infrastructure and services	Virtualization; cloud computing/infrastructure as a service; cloud integration; serverless computing; systems analysis and design; network design; systems auditing; wireless; telecommunications/VoIP (Voice over Internet Protocol); database administration; data centers
Security	IT security planning and management; BYOD; governance, risk, and compliance; cybersecurity
Applications	Customer-facing application development; mobile app development; web development; open source; portal technologies; cloud computing; user experience; legacy systems integration; interface design; content management systems
Internet	Social media; customer-facing web applications; mobile apps; search engine optimization; artificial intelligence; web mining; Internet of Things
Business analytics/ data science	Business intelligence; advanced analytics; data warehousing; data mining; unstructured data analysis; Big Data

Source: Based on Broom (2016), Florentine (2015), Pratt (2015), Leung (2009).

Perhaps now you can see why IS professionals are so valuable to their organizations. These individuals have a solid, integrated foundation in technical, business, and systems knowledge and skills. Perhaps most important, they also have the social skills to understand how to work well with and motivate others. It is these core competencies that continue to make IS professionals valuable employees.

Given how important technology is, what does this mean for your career? Technology is being used to radically change how business is conducted—from the way products and services are produced, distributed, and accounted for to the ways they are marketed and sold. Whether you are majoring in information systems, finance, accounting, operations management, human resource management, business law, or marketing, knowledge of technology is critical to a successful career in business.

FINDING QUALIFIED PERSONNEL. Unfortunately, given the increased sophistication of modern information systems, organizations can often have a difficult time finding qualified personnel, and attracting the right people with the right skills is not possible in some areas. Consequently, many technology-focused organizations tend to cluster in areas where talented workers are available. Such areas are often characterized by a high quality of life for the people living there, and it is no surprise that many companies in the IT sector within the United States are headquartered in Silicon Valley, California; Boston, Massachusetts; Austin, Texas; or Seattle, Washington. With increasing globalization, other regions throughout the world are boasting about their highly skilled personnel. One such example is the Indian city of Bangalore, where, more than a century ago, Maharajas started to lure talented technology-oriented people to the region, building a world-class human resource infrastructure that attracted companies from around the world. In other areas, organizations may have to find creative ways to attract and retain people, such as by offering favorable benefits packages that include educational grants or expense-matching programs to encourage employees to improve their education and skills. Other human resource policies, such as telecommuting, flextime, and creative benefit packages, can also help to attract and retain the best employees.

YOU—THE USER. Clearly, the field of information systems offers a wide variety of interesting career choices, and you will likely find a career that offers a host of opportunities for lifelong learning and advancement. Yet understanding fundamental concepts related to information systems will be critical in almost any career as well as in your private life. In almost any



SECURITY MATTERS

Ransomware

As more and more of our business and professional lives are lived online, the security of our personal and business data has become increasingly important. In recent years, attackers have gotten more sophisticated and have shifted tactics. A new tactic is the distribution of ransomware, a novel approach to extracting money from victims. Ransomware refers to a type of virus that, once it has infected a victim's system or network, encrypts the data it finds in place in a format that renders them impossible for the victim to access. The attacker then demands a ransom payment in return for releasing the decryption keys that can be used to access and recover the data (though many victims have painfully experienced that paying the ransom does not guarantee regaining access to the data). Victims range from individuals, who may stand to lose years of family photos or personal records, to businesses large and small, who may lose customer records, financial data, intellectual property, or other valuable data. The malicious software usually arrives as an attachment to a spam e-mail or is downloaded in the guise of a video or other content from a website.

Until recently, attackers have largely focused on individuals and small businesses and demanded relatively small ransoms that are affordable enough that the victim will seriously consider paying the ransom. The average payment demanded in 2015 was just US$300 according to security firm Symantec. By early 2016, however, there has been an apparent shift to larger, higher-value organizations. Hollywood Presbyterian Medical Center in Los Angeles was forced to pay a US$17,000 ransom after its network was broadly infected with ransomware. Another hospital in Ottawa decided not to pay the ransom demanded of it and was able to restore data from backups. Security researchers warn that attackers are increasingly not just attempting to lock down the data but are threatening to release the data to the public if not paid. For organizations dealing with sensitive or personally identifiable information, lawsuits and reporting requirements can make such public release an even more costly threat than simply losing access to their data.

With threats like ransomware becoming increasingly prevalent, individuals and organizations will need to increase their security awareness and vigilance to better avoid potential infections and also improve their backup and disaster recovery preparations in order to become more resilient to such attacks.

Based on:

Colwey, S., & Stack, L. (2016, February 18). Los Angeles hospital pays hackers $17,000 after attack. *The New York Times.* Retrieved April 14, 2016, from http://www.nytimes.com/2016/02/19/business/los-angeles-hospital-pays-hackers-17000-after-attack.html

Pilieci, V. (2016, March 16). Ottawa Hospital hit with ransomware, information on four computers locked down. *Ottawa Citizen.* Retrieved April 14, 2016, from http://ottawacitizen.com/news/local-news/ottawa-hospital-hit-with-ransomware-information-on-four-computers-locked-down

Savage, K., Coogan, P., & Lau, H. (2015, August 6). The evolution of ransomware. *Symantec.com.* Retrieved April 14, 2016, from http://www.symantec.com/content/en/us/enterprise/media/security_response/whitepapers/the-evolution-of-ransomware.pdf

business-related field, you will be extensively using information systems, and you will likely be involved in various information systems–related decisions within your organization. Understanding what information systems are capable of doing (as well as what they cannot do), being able to communicate with the "techies," and being able to make educated IS-related decisions are likely to set you apart from your competition. Especially in smaller organizations (that may not have dedicated IS departments), you are likely to be involved in IS-related investment decisions, and lacking a basic understanding of fundamental issues associated with topics such as IS infrastructure, systems analysis and design, or information systems security will put you at the mercy of outside consultants or (worse yet) vendors who are likely to act out of their own interests, often trying to sell you their "technology of the week/month/year."

In addition, as you have undoubtedly noticed, you are facing a number of IS-related decisions in your private life. Examples of such decisions abound; for example, you may face the question of what mobile phone to purchase next: an iPhone, a phone using some version of the Android operating system, or a phone sporting Microsoft's Windows 10 Mobile operating system. Such decisions are likely to include your own preferences or influence by your peers, but there are a number of critical differences in terms of privacy, security, available apps, and the like. Likewise, you may face the problem of how to best secure your wireless network at home or may wonder how to best keep your various files in sync across different computers or mobile devices.

Finally, you may have a great idea for a new product or service and want to launch a startup. Having the idea is but the first step, and you will soon realize the role of information systems in

all phases of the process of bringing your idea to the market. Thus, understanding how information systems can fuel the development and commercialization of your idea is crucial. Throughout this text, we will touch on those issues and hope that you will gain valuable knowledge to understand the trade-offs involved when selecting new information systems.

Organizations: The Context of Information Systems

We have talked about data versus information, the technology side of information systems, and the people side of information systems. Information systems do not exist in a vacuum; they are built and/or used within a certain context. Organizations use information systems to become more productive and profitable, to gain competitive advantage, to reach more customers, or to improve customer service. This holds true for all types of organizations—professional, social, religious, educational, and governmental—and for all types of industries—medical, legal, manufacturing, and so on. In fact, the U.S. Internal Revenue Service launched its own website for the reasons just described (Figure 1.15). The website was so popular that approximately 220,000 users visited it during the first 24 hours and more than 1 million visited it in its first week—even before the web address for the site was officially announced. Today, popular websites like Facebook.com and WSJ.com receive millions of visitors every day.

TYPES OF INFORMATION SYSTEMS. Throughout this text, we explore various types of information systems commonly used in organizations. It makes sense, however, for us to describe briefly a few of the various types of systems used so that you will better understand what we mean by the term *information system* as we use it throughout the rest of the book. Table 1.5 provides a list of the major categories of information systems used in organizations.

Topping the list in the table are some of the more traditional, major categories that are used to describe information systems. For example, not only are **transaction processing systems (TPS)** used by a broad range of organizations to process customer transactions more efficiently, these systems also generate a tremendous amount of data that can be used by the organization to learn about customers or ever-changing product trends. Your local grocery store uses a TPS at the checkout that scans bar codes on products; as this occurs, many stores will print discount coupons on the backs of receipts for products related to current purchases. Every hour, online retailer Amazon.com's website processes thousands of transactions from around the world. This massive amount of data is fed into large data warehouses and is then analyzed to provide purchase recommendations to future customers. In addition, TPS data are sorted and organized to support a broad range of managerial decision making using a variety of systems; the most common of these is generally referred to as **management information systems**. TPS data also provide input into a variety of other information systems within organizations, including *decision*

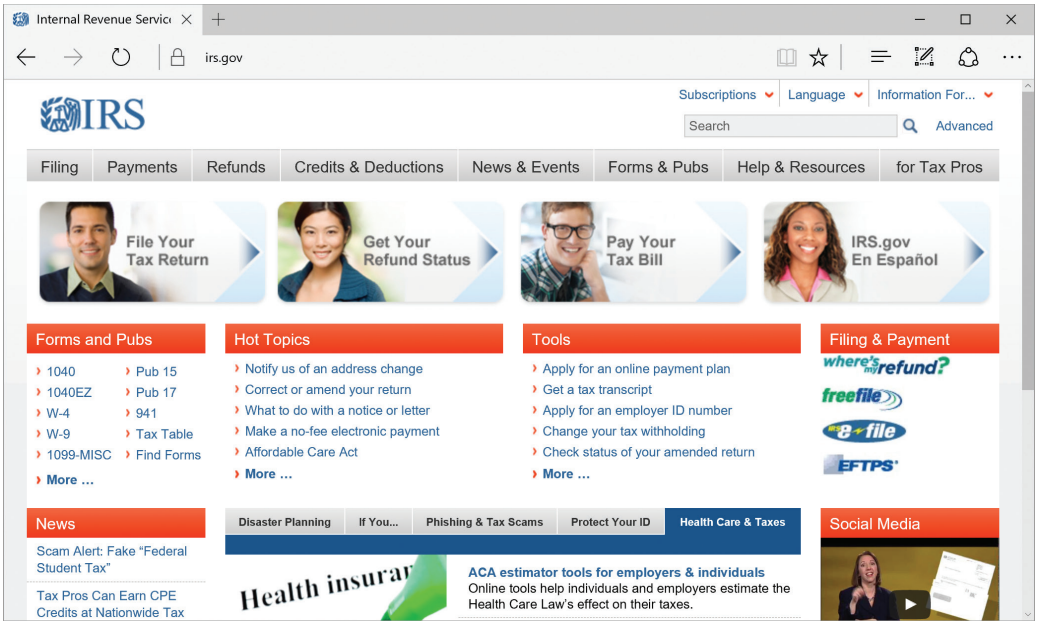


FIGURE 1.15

Website of the U.S. Department of the Treasury, Internal Revenue Service, http://www.irs.gov.

Source: Courtesy of the United States Department of the Treasury.

TABLE 1.5 Categories of Information Systems Used in Organizations

Category of System	Purpose	Sample Application(s)
Transaction processing system	Process day-to-day business event data at the operational level of the organization	Grocery store checkout cash register with connection to network, student registration system
Management information system	Produce detailed information to help manage a firm or part of a firm	Inventory management and planning system, student enrollment management
Decision support system	Provide analysis tools and access to databases in order to support quantitative decision making	Product demand forecasting system, loan and investment analysis
Intelligent system	Emulate or enhance human capabilities	Automated system for analyzing bank loan applications, self-driving car
Business intelligence system	Analyze Big Data to better understand various aspects of a business	Online analytical processing (OLAP) system
Office automation system (personal productivity software)	Support a wide range of predefined day-to-day work activities of individuals and small groups	Word processor, spreadsheet, presentation software, e-mail client
Collaboration system	Enable people to communicate, collaborate, and coordinate with each other	E-mail system with automated, shared calendar
Knowledge management system	Enable the generation, storage, sharing, and management of knowledge assets	Knowledge portal for finding answers to common questions
Social software	Facilitate collaboration and knowledge sharing	Social network, connecting colleagues and friends
Geographic information system	Create, store, analyze, and manage geographically referenced data	Site selection for new shopping mall
Functional area information system	Support the activities within a specific functional area of the firm	Planning system for personnel training and work assignments
Customer relationship management system	Support interaction between the firm and its customers	Sales force automation, lead generation
Enterprise resource planning system	Support and integrate all facets of the business, including planning, manufacturing, sales, marketing, and so on	Financial, operations, and human resource management
Supply chain management system	Support the coordination of suppliers, product or service production, and distribution	Procurement planning
Electronic commerce system	Enable customers to buy goods and services from a firm's website	Amazon.com, eBay.com, Nordstrom.com
Mobile app	Perform a particular, well-defined function, typically on a mobile device	Instagram, Snapchat, WhatsApp, Facebook app

support systems, intelligent systems, business intelligence systems, knowledge management systems, social software, geographic information systems, and *functional area information systems*. Five to 10 years ago, it would have been typical to see systems that fell cleanly into one of these categories. Today, many organizations have replaced standalone systems with *enterprise systems* that span the entire organization. Likewise, with **internetworking**—connecting host computers and their networks together to form even larger networks like the Internet—and **systems integration**—connecting separate information systems and data to improve business processes

and decision making—it is difficult to say that any given information system fits into only one of these categories (e.g., that a system is a management information system only and nothing else). In addition, many of these systems are not housed within organizations anymore but are located "in the cloud" and accessed via the users' browsers when needed. Modern-day information systems tend to span several of these categories of information systems, helping not only to collect data from throughout the firm and from customers but also to integrate data from diverse sources and present them to busy decision makers along with tools to manipulate and analyze those data. *Customer relationship management*, *supply chain management*, and *enterprise resource planning* systems are good examples of these types of systems that encompass many features and types of data and cannot easily be categorized.

Office automation systems such as Microsoft Office and the OpenOffice.org Productivity Suite provide word processing, spreadsheet, and other personal productivity tools, enabling knowledge workers to accomplish their tasks; *collaboration systems*, such as Microsoft's Exchange/Outlook, Lotus Notes, or Google Apps, provide people with e-mail, automated calendaring, and online, threaded discussions, enabling close collaboration with others, regardless of their location.

Systems for *electronic commerce* (or *e-commerce*), such as corporate websites, are also popular and important. These systems enable (1) consumers to find information about and to purchase goods and services from each other and from business firms and (2) business firms to electronically exchange products, services, and data. In Chapter 4, "Enabling Business-to-Consumer Electronic Commerce," we talk about different forms of electronic commerce involving the end consumer; in Chapter 8, "Strengthening Business-to-Business Relationships via Supply Chain and Customer Relationship Management," we discuss how organizations use the Internet to enable or facilitate business-to-business transactions.

While many modern-day information systems span several of these IS categories or integrate different types of systems, it is still useful to understand these categories. Doing so enables you to better understand the myriad approaches, goals, features, and functions of modern information systems.

We have talked about each of the parts of our definition of information systems, and we have talked about different types of information systems. In the next section, we focus on how information systems can be managed within organizations.

ORGANIZING THE IS FUNCTION. Old-school IS personnel believed that they owned and controlled the computing resources, that they knew better than users did, and that they should tell users what they could and could not do with the computing resources; in addition, early IS departments typically had huge project backlogs, and IS personnel would often deliver systems that were over budget, were completed much too late, were difficult to use, and did not always work well. The increasing pervasiveness of technology in businesses and societies has led to a shifting mindset about information systems within organizations. Increasingly fast-paced competition is forcing businesses to regard IS as an enabler for streamlining business processes, providing better customer service, and better connecting and collaborating with various stakeholders inside and outside the organization. Many organizations, for example, have realized that some of the best ideas for solving business problems come from the employees using the system; as a result, personnel within many IS units have taken on more of a consulting relationship with their users, helping the users solve problems, implement ideas, and be more productive. IS personnel are increasingly reaching out to their internal customers and proactively seek their input and needs rather than waiting for customers to come in with systems complaints. They modify the systems at a moment's notice just to meet customer needs quickly and effectively. They celebrate the customers' new systems ideas rather than putting up roadblocks and giving reasons that the new ideas cannot or will not work. They fundamentally believe that the customers own the technology and the information and that the technology and information are there for the customers, not for the systems personnel. They create help desks, hotlines, information centers, and training centers to support customers. These service-oriented IS units structure the IS function so that it can better serve the customer.

The implications of this new service mentality for the IS function are staggering. It is simply amazing how unproductive a company can be when the IS personnel and other people within the firm are at odds with one another. On the other hand, it is even more amazing how productive and enjoyable work can be when people in the IS function work hand in hand with people